

Coalition Deception and Defensive Adaptation in a Social-Deduction Benchmark

Isaac Lin

Xi Chen

Abstract

Defenses against coordinated testimony need evaluation against adversaries trained to exploit their decisions. We study this question in a social-deduction benchmark with structured claims, exact truth labels, partial observations, and two impostors sharing a reinforcement-learning policy. A completed 240-job replication uses repaired episode collection, independent evaluation, matched private information, and seed-level inference. Across ten seeds at each of three population sizes, meeting-only learning increases false ejection and aligned voting while directly false claims decrease. Spatial adaptation depends on the opponent and the outcome: in the five-crew, two-impostor multi-round game, coalition victory falls from 0.967 to 0.360 after defender training and returns to 0.851 after coalition retraining, while innocent ejections remain frequent. Ten-generation crossplay shows renewed gains against the latest opponent, without establishing convergence or a permanent cycle. A hypothesis-based defender incurs a truthful-testimony accuracy cost and is vulnerable to a targeted coalition: false ejection rises from 0.126 to 0.422 at five plus two players. Supplemental ballot diagnostics implicate honest abstention under fixed recorded votes. A 540,000-episode mechanism study finds increased alibi errors under credibility weighting despite coalition weights below their uniform shares. A separate Gaussian surrogate characterizes weight amplification under explicit assumptions. Dependence-aware defense has mixed effects, including increased innocent ejections in the learning loop. These results distinguish coalition victory, voting harm, and credibility mechanisms, and bound what finite adaptive evaluations establish about robustness.

1 Introduction

In a social-deduction game, players combine private observations and public testimony to decide whom to eject. Agreement between speakers can reflect independent evidence or coordination within a hidden coalition. A defense that handles one scripted form of corroboration may face a different problem when the coalition is trained against its decisions.

We study an Among Us-inspired benchmark in which two impostors share a policy and objective. Structured claims provide explicit truth and support labels, while the same meeting engine accepts scripted testimony, learned coalitions, and scripted or learned defenders. Our central question is whether defensive performance transfers from specified attacks to adver-

saries trained against the defense, and how this interaction changes over repeated response training.

All empirical game results in this manuscript use the completed corrected replication. An implementation audit identified incomplete spatial trajectories, duration-dependent evaluation selection, unequal private-history retention, and defects in compatibility and hypothesis scoring. These were repaired before the 240-job replication; historical learned outcomes are not pooled with it. Final inclusion checks accepted every planned job and all 303 contrasts. The post-audit analysis specification is not a prospective preregistration of the original research. The separate mechanism and ballot studies are descriptive supplements.

The findings distinguish several claims that outcome curves alone can conflate. Meeting-only training increases harmful voting alongside greater coalition alignment, even as directly false claims fall. The repaired hypothesis rule improves some scripted-attack outcomes but incurs a large truthful-testimony cost. Spatial defender training changes incident opportunities and voting behavior; its benefit depends on whether success means preventing coalition victory or preventing innocent ejection. Repeated training creates renewed gains against the latest opponent, but finite crossplay does not identify asymptotic dynamics. Adaptive attacks against the hypothesis rule expose a substantial vulnerability, whereas dependence-aware defenses have mixed effects across harms and opponents.

A Gaussian report model separately supplies exact agreement probabilities and a deterministic mean-field surrogate for corroboration-based weight amplification. Its assumptions and finite-sample limitations are explicit. Direct measurement in the game instead finds alibi errors without excessive total coalition credibility weight, and targeted score interventions have attack-dependent costs. The benchmark therefore supports a comparison of mechanisms and finite responses, rather than a universal ranking of defenses.

2 Related work and contribution

Strategic information transmission was formalized by Crawford and Sobel’s model of cheap talk [1], where conflicting preferences produce coarse communication equilibria. Lewis’s signaling game provides the cooperative counterpart in which shared objectives support emergent conventions [2]. Reinforcement-learning studies such as EGG [3] and *Cheap Talking Algorithms* [4] bring these signaling problems into trainable sender-receiver systems. Work on multiple senders studies how coalition size and disagreement affect information aggregation [5], [6].

Related defenses include robust aggregation, Byzantine-resilient multi-agent reinforcement learning, and peer-prediction mechanisms [6], [7], [8]. Their guarantees have explicit assumptions; arbitrary Byzantine behavior is not equivalent to a fixed scripted attack. Results on the geometric median also identify limits to strategyproofness [9]. Social-deduction systems such as *Hidden Agenda* and language-model studies provide a second line of work on learned deception, hidden roles, and strategic discussion [10], [11]. Work on adversarial communication, multi-agent credit assignment, emergent cooperation under uncertain incentives, and secret collusion provides adjacent models [12], [13], [14], [15].

Evaluating a defense with attacks adapted to it is an established requirement in adversarial robustness [16]. Learning approximate responses to policies or policy populations is also established in multi-agent reinforcement learning [17]. Our contribution is an empirical comparison within a structured social-deduction benchmark, rather than the general observation that a defense needs adaptive evaluation.

The comparison uses common defenses within each environment configuration and contrasts scripted with trained adversaries. The meeting-only and spatial games differ, so comparisons across those configurations are not controlled changes of attack policy alone. Four further elements make the comparison possible and interpretable.

1. **An interpretable environment.** Private evidence, public claims, response actions, votes, roles, and state transitions are recorded explicitly. Structured claims allow truth and support labels to be computed without interpreting natural-language dialogue, and the same meeting engine accepts scripted coalitions with a fixed dependence structure and learned coalitions that choose their own.
2. **A mechanism-level comparison of aggregators.** The engine can use order-statistic rules, credibility weighting, a dependence-aware discount, or a heuristic hypothesis-scoring rule, and every rule’s realized weights can be measured. This is what makes it possible to say *through which channel* a scripted attack acts, and therefore whether a learned coalition uses the same channel.
3. **A staged and a long-horizon adaptation study.** The $C0 \rightarrow D1 \rightarrow C1$ protocol separates initial coalition learning, defender response, and coalition adaptation; its continuation for ten generations, with cross-generation play and policy-distance tracking, describes the finite-horizon sequence of approximate responses trained from fresh initializations. Controls for the coalition’s coordination channel, the training budget, and the reward specification test sensitivity to the selected information, budget and reward settings.
4. **An analytical model** of credibility weighting under correlated testimony, with exact agreement probabilities and a deterministic surrogate checked alongside Monte Carlo simulations and the game’s measured credibility weights, so that the one failure mode the model describes, corroboration capture, can be distinguished from the ones the benchmark exhibits.

3 Environment

The corrected implementation is used throughout the game experiments. Appendix C records its source identity and validation; the historical implementation is preserved separately.

3.1 Game structure

The benchmark contains two hidden impostors and a variable number of crew members. The main crew sizes are 3, 5, and 7, producing 3+2, 5+2, and 7+2 players. Every experiment uses one of two configurations of the same engine (Table 1).

Table 1. The two configurations. Both use the same claim, response, vote, truth-labeling and transition code; roles are re-randomized every episode, the reporter announces the incident time, and roles are not revealed on ejection.

	Meeting-only game (F1, F2)	Spatial short game (F3 to F5, F7, F8, controls)
Rooms	Hub, Electrical, Navigation, Medbay	same
Edges	Hub–Electrical, Hub–Navigation, Hub–Medbay	the same three plus Electrical–Navigation
Evidence	positions sampled by a random walk on the map (stay probability 0.35), 6 free-play turns	positions produced by the agents’ movement actions, 8 free-play turns, two tasks per agent
Incident	one incident placed by the episode process	created by an impostor action with one crew target and no other living crew in the room
Coalition channel	none	learned private symbol channel, four symbols
Meeting	claim round, response round, vote (plurality, ties do not eject)	same
Rounds	one meeting	one meeting (F3) or up to eight rounds with terminal win conditions (F4 onward)

The meeting-only game generates evidence exogenously and then runs a structured meeting, which makes the scripted rule comparison of F2 easy to reproduce with labeled evidence. The spatial short game produces evidence through movement, tasks, and an isolation-gated incident action, and adds a learned low-bandwidth private channel between the two impostors. The spatial map is the three-edge star of the meeting-only game plus one extra edge between Electrical and Navigation, so a player can move between those rooms without passing through the Hub.

3.2 Information and actions

Each player receives a private view containing its own evidence and a public transcript containing the claims and responses already made. Roles remain hidden in the public transcript. Actions are factorized into legal heads: movement between connected rooms; a task, observe, or report action; an incident action for an impostor when the local conditions permit it; a structured claim about a room, time, or observed player, with a polarity; a response to another claim (accuse, defend, agree, contradict, or neutral); a private symbol in the spatial game; and a vote for a player or skip.

Claims are structured symbolic objects. Claims have a truth/support label (true, unsupported, false, or contradictory) relative to the recorded state and the speaker’s evidence, which provides exact measurement of strategic testimony. No-information is a separate claim action, not a fifth truth label. The scripted policies reconstruct the observer’s full within-round co-location history, including negative observations, to check claims. Learned spatial actors receive the public transcript and a time-indexed private-history block containing observation validity, their own room, and co-located agents. This retains the same positive and negative positional evidence without exposing unseen rooms or future positions.

The two impostors know one another and share a policy. An identity feature marks the partner in each impostor’s observation so that the two copies of the shared policy can specialize within the coalition. In the spatial game the coalition also has the learned symbol channel. Both the identity feature and the channel are ablated in Section 6.5.

3.3 Scripted policies

The scripted **truthful crew** moves toward its task rooms, reports an incident when it finds the marker, and in the meeting states its most informative *supported* claim: among the claims its own trajectory entitles it to make, it prefers claims about other agents over claims about itself, positive sightings over negative ones, and claims at or adjacent to the incident time and in the incident room. It speaks the claim’s own polarity, so a crew member that did not see an agent in its room says so rather than asserting the opposite. It accuses the most suspicious agent when its suspicion exceeds a threshold and votes the same way, skipping otherwise. Its suspicion is formed by the aggregation rule of Section 3.4.

Four scripted **coalitions** define F2. The *truthful* coalition uses supported claims and the crew’s suspicion rule, while filtering some incriminating claims and avoiding coalition members in voting. Truthful claim contents therefore do not define a fully cooperative electorate.

For the *lone liar*, the creator claims to have been elsewhere, accuses a selected innocent and votes for that target; the partner speaks truthfully without defending the creator. In the *coordinated alibi*, the creator makes a self-location claim away from the scene, while the partner claims to have seen the creator away from it. The creator accuses an innocent, the partner defends the creator, and both vote for the selected innocent. Without a channel, their asserted rooms are not guaranteed to agree. The *coordinated framer* has both members claim to have seen the selected innocent in the incident room at the incident time, then accuse and vote for that player. Those framing claims are not necessarily truthful. These are distinct scripts, not pure manipulations of one agreement variable.

Scripted spatial coalitions use their own last sightings, public alive status, and legitimately known partner and marker information when choosing movement. They do not inspect unseen neighboring occupancy. F2 instead uses exogenous evidence, so movement is not a learned choice.

3.4 Aggregation rules

A crew voter forms a suspicion score over the other agents from two sources, its own grounded evidence and testimony, and the aggregation rule decides how testimony is combined:

$$\text{suspicion}_v(j) = \text{own}_v(j) + \text{AGG}_s [\text{signal}_v(s, j)],$$

where $\text{signal}_v(s, j)$ is how much speaker s 's claims and responses implicate candidate j as judged by voter v . A claim that places j in the incident room at the incident time raises j 's suspicion; a claim by another speaker that places j elsewhere lowers it (the *alibi-witness* term); an accusation raises it; and a speaker whose claim the voter personally knows to be false is implicated only itself, by a large *caught-lying* penalty that sits in the liar's own row. The voter's own evidence marks agents it saw at the scene, clears agents it saw elsewhere, and treats agents it cannot place as suspects by default.

- **Mean:** the sum of the per-speaker rows.
- **Median and trimmed mean:** the per-candidate median, or the mean after discarding the extreme row on each side, rescaled to the sum's scale.
- **Soft and sharp credibility:** a weighted sum in which a speaker's weight is the softmax of its credibility: -3 if the voter catches a false claim, otherwise -2 for a self-contradictory claim set, otherwise the count of other informative speakers with jointly satisfiable claims; the temperatures are 1.5 (soft) and 0.15 (sharp).
- **Dependence-aware credibility:** soft credibility whose weights are multiplied by $e^{-\lambda \text{ex}_s}$, where ex_s is how much more s agrees with its closest partner than with everyone on average. Agreement is pooled from the public record over four channels (joint satisfiability of claims, one speaker clearing the other of the scene, defend or agree responses, and the same non-skip vote) over a sliding window of three meetings; $\lambda = 4$. A **dependence-weighted vote** applies the same discount to ballots instead of to testimony. This ballot-level rule remains active when a learned crew replaces the scripted testimony aggregator. **Hypothesis scoring.** The rule enumerates size-two coalitions

and averages over declared honesty gates, treating each coalition speaker’s informative claims as accepted with probability one half or ignored otherwise. It excludes publicly known crew, including murdered victims inferred from public death and ejection records. Accepted claim constraints and private positive/negative observations are solved jointly, with public inactivity respected. It averages feasibility indicators over the declared honesty gates and auxiliary creator/time assignments, normalizes positive scores, and gives infeasible candidates zero weight; an all-zero view skips when legal. Membership, creator score, own evidence, and agent index provide exact lexicographic tie breaking. A vote requires leading membership score above one half; ties are resolved by creator score, own evidence, and finally agent index. This remains a feasibility heuristic, not a calibrated posterior for the full game-generating process. The shared solver now checks all emitted constraints using propagation and backtracking; some negative transition claims still use a documented necessary-constraint relaxation.

The credibility rules use exactly the same information as the mean; they differ only in how much each speaker’s row counts.

4 Learning procedure

4.1 PPO policy

The learned agents use a masked actor-critic policy trained with Proximal Policy Optimization (PPO) [18]. The actor maps an observation to a probability distribution over each action head; legal-action masks remove actions unavailable to that observer [19]. In the corrected hidden-partner intervention, incident masks allow role-blind attempts that can resolve as a legal no-op under hidden game rules, so mask entries do not reveal the partner’s role. The critic maps the observation to a scalar value estimate. The clipped actor objective is

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right],$$

where $\hat{A}_t$ is the estimated advantage and $r_t(\theta)$ the new-to-old action probability ratio. Both actor and critic are two-layer networks with 128 hidden units; the learning rate is 3×10^{-4} , $\gamma = 1$ for terminal episodic rewards, generalized advantage estimation uses $\lambda = 0.95$, and each update runs up to eight PPO epochs over four minibatches with an entropy coefficient annealed from 0.02 to 0.005. PPO can stop an update early when approximate KL exceeds 0.075 (1.5 times the target KL of 0.05). The policy is non-recurrent: the meeting observation contains the public transcript, so the visible history is available at each meeting, but long-horizon memory across rounds is a limitation of the multi-round study. The standard stage budget is 400 updates; defender-budget controls use 800 or 1,600. F1 collects 64 completed episodes per update across 64 parallel environments. Spatial runs collect 32 completed episodes across 32 environments; the collector waits for the complete batch before assigning terminal targets. No unfinished trajectory is finalized as terminal. Evaluation samples actions from the policy, using a copied policy with an independently seeded action generator.

4.2 Reward design

The reward is terminal. Directly rewarding “lie more”, “agree with your partner”, or “protect the killer” would program the behavior that the experiment is supposed to measure.

In F1 the coalition uses the incident-gated `false_ejection` objective: an innocent ejection during a real incident gives the coalition +1, while a skip, a coalition ejection, or a meeting without an incident gives −1. Only the coalition is trained; the crew follows the scripted soft-credibility policy. F2 is fully scripted.

In the single-meeting response cycle (F3) and its budget control every stage uses the `balanced_ejection` objective, which is zero-sum: the coalition receives +1 for an innocent ejection, 0 for a skip, and −1 for a coalition ejection or for a meeting without an incident, and the crew receives the negative of the coalition’s score. The trained side optimizes its own score while the opposing checkpoint is fixed. The single-meeting multi-generation loop (F5 and F7) keeps `balanced_ejection` for the coalition stages and trains the crew stages on `meeting_ejection`, under which only a real incident followed by an ejection is scored (a coalition ejection is +1 for the crew, an innocent ejection −1, skips and incident-free meetings 0): the defender is trained on the quality of real meeting decisions, while the coalition is also penalized for failing to create an incident. The two constructions are compared directly by the reward control of Section 4.4.

For the multi-round game (F4 onward) the objective is terminal survival: the coalition wins when, after a meeting vote, the number of living impostors is at least the number of living crew; the crew wins when the coalition is eliminated or the round cap is reached without coalition parity. The reward control also replaces survival by a framing-count objective (`false_ejection`: the coalition’s terminal reward counts innocents ejected in rounds with a real incident; the crew’s is its negative).

4.3 Staged and multi-generation training

The response-cycle experiments use frozen checkpoints:

C0: coalition learns against the scripted soft-credibility crew

↓ freeze C0

D1: crew learns against frozen C0

↓ freeze D1

C1: coalition learns against frozen D1

This order asks how a defender policy responds to a known learned coalition and how much of that response survives when the coalition is trained again. It is a sequential response process; simultaneous co-evolution is a separate condition.

Each new stage initializes a fresh actor and critic; learner weights are not warm-started from the preceding same-side policy. The frozen opponent is loaded from its checkpoint. Thus policy distances include effects of independent training and initialization.

The cycle is continued as $C0 \rightarrow D1 \rightarrow C1 \rightarrow D2 \rightarrow C2 \rightarrow \dots \rightarrow Dk \rightarrow Ck$. Every stage trains

one side for the same budget against the other side’s frozen checkpoint; D0 is the scripted soft-credibility crew, and stage s of a run with seed σ uses training seed $\sigma + 10000s$, with matched per-stage seed offsets and standard budgets. Single-meeting reward definitions still differ between the staged cycle and the default multi-generation loop. After every stage the frozen match-up is evaluated on 1,000 episodes, both sides’ policies are compared with their predecessors on a fixed scripted probe set whose size depends on the configuration (Kullback–Leibler and Jensen–Shannon divergences summed over masked action heads and averaged over probe rows, a behavioral Jensen–Shannon divergence between vote distributions in canonical categories, and the fraction of greedy actions that differ), and after the last stage every C_i is played against every D_j (500 episodes per cell). Ten training seeds are run for $k = 10$ at 5+2 in both regimes and at 3+2 in the multi-round game; five seeds are run for $k = 3$ at 7+2 in the multi-round game.

4.4 Controls

Coordination ablation. The C0 stage of the spatial game (5+2 players, single meeting and multi-round) is trained under four information conditions: the default (partner identity known, private channel on), no channel, partner identity hidden, and neither. Five seeds each.

Defender budget. The single-meeting cycle at 5+2 is repeated with the defender stage trained for 800 and 1,600 updates instead of 400, and the multi-round cycle with 800, while C0 and C1 keep 400 updates; five seeds, the same seeds as the reference runs.

Reward specification. The single-meeting loop at 5+2 is repeated for four generations with one objective for both sides, either `balanced_ejection` (the F3 construction) or `meeting_ejection`, so that either the crew’s reward for incident-free meetings is introduced or the coalition’s incident-free penalty is removed; both objectives give zero reward for a skip following an incident; the multi-round loop at 5+2 is repeated for two generations with the framing-count objective in place of survival. Five seeds each, the same seeds as the reference loops, so every comparison is paired by seed.

5 Evaluation protocol

All 240 planned jobs, containing 1,255 learner stages, completed and passed source, job, configuration, checkpoint, and raw-record checks. The independent unit is a training seed for learned experiments and an independently seeded evaluation batch for scripted experiments. Episodes and generations are repeated observations within that unit. No seed was removed or stopped because of its outcome.

The primary single-meeting outcome is false-ejection probability, together with creator-ejection, no-ejection, and incident rates. Creator ejection and creator survival both require a real incident; they are not unconditional complements. Conditional rates are computed within each seed and averaged only over seeds with nonzero denominators, with contributing counts retained. For multi-round play we report coalition victory, probability of any inno-

cent ejection during the game, and mean innocent-ejection count. Terminal-meeting false ejection and pooled false-ejections per meeting remain separate descriptive quantities. Every completed meeting, including incident-free meetings, contributes to whole-game records. Dependence statistics include the complete electorate before ejection and count each historical meeting once.

F1 uses ten seeds per population, 400 updates, and 400 holdout episodes at the genuinely untrained and final checkpoints. Monitored curves use a separate evaluation stream. F2 uses ten independent replicates and 1,000 episodes per rule/condition at each size. F3 and F4 use ten seeds per population and 1,000 final episodes for each of four pairings. The main multi-generation loops use ten seeds for 5+2 single- and multi-round play and 3+2 multi-round play through ten generations; the 7+2 multi-round loop uses five seeds through three generations. Each stage has 1,000 evaluation episodes; full crossplay uses 500 per cell. The defended loop uses ten seeds and four generations. Adaptive dependence comparisons use ten seeds at 5+2 and five at 3+2; adaptive hypothesis comparisons use ten at 5+2 and five at 7+2. Static spatial and control studies use five seeds. All seed identities, budgets, metrics, and signed comparison expressions are supplied in the versioned protocols.

The final evidence stream starts at 1987654321 plus the run seed, separately from monitored evaluation at 987654321 plus the run seed. Evaluations finish predetermined episode indices rather than selecting the fastest completions, and do not advance training randomness. F2 streams are condition/rule-specific; pairing is by replicate ID and does not imply that its full transcripts are held fixed. Supplemental matched-evidence studies state their stronger matching separately.

For each of 303 specified contrasts in 15 families, identical seed IDs are paired before subtraction. We report the mean effect, sample SD, a pointwise 95% percentile interval from 20,000 resamples of complete paired seed differences, and a two-sided exact sign-flip test of the paired mean. The test preserves effect magnitudes and assumes sign symmetry/exchangeability under the null; pairing alone does not ensure this assumption. Holm correction applies within each complete declared family, not across the entire paper. All planned contrasts, including unfavorable and nonsignificant results, are distributed with the analysis.

With ten seeds the minimum raw two-sided p-value is 0.001953125; with five it is 0.0625. Consequently a five-seed test cannot reject at raw 0.05, and a family of at least 26 ten-seed tests cannot pass Holm's first threshold at 0.05. The F4 family contains 27 contrasts, so its smallest adjusted p-value is 0.0527. This resolution limit was retained rather than changing families after observing outcomes. Confidence intervals are pointwise, not simultaneous; failure to reject does not establish equivalence. Unless a paired interval or adjusted p-value is stated, numerical comparisons are descriptive means and sample SDs.

The 540,000-episode mechanism and 67,500-episode ballot studies are post-audit descriptive supplements. The former has ten replicates per population and matches initial evidence across nine rule/variant cells within each condition; responses and votes may change. The latter evaluates retained hypothesis-attack checkpoints across 135 cells, with 500 episodes per cell, ten seeds at 5+2 and five at 7+2. It retains complete pre-ejection ballots and honest score views. Removing ballots in that diagnostic holds other recorded votes fixed,

not behavior under a different game.

For each multi-generation crossplay matrix M , the exploratory finite-pool gap is $G_k = \max_i M_{ik} - \min_j M_{kj}$, using all observed policies, including policies trained after generation k . Simultaneous Hoeffding bands over the finite matrix account for evaluation uncertainty and selection of empirical extrema; independent episode draws within each cell are assumed, while shared seeds across cells are allowed. This is a retrospective policy-pool comparison, not true exploitability or a stopping rule. Neither gaps, policy distances, nonsignificant trends, nor observed alternation prove an equilibrium or asymptotic cycle.

6 Results

6.1 F1: learning against a fixed scripted defense

F1: monitored learning curves and independent holdout endpoints

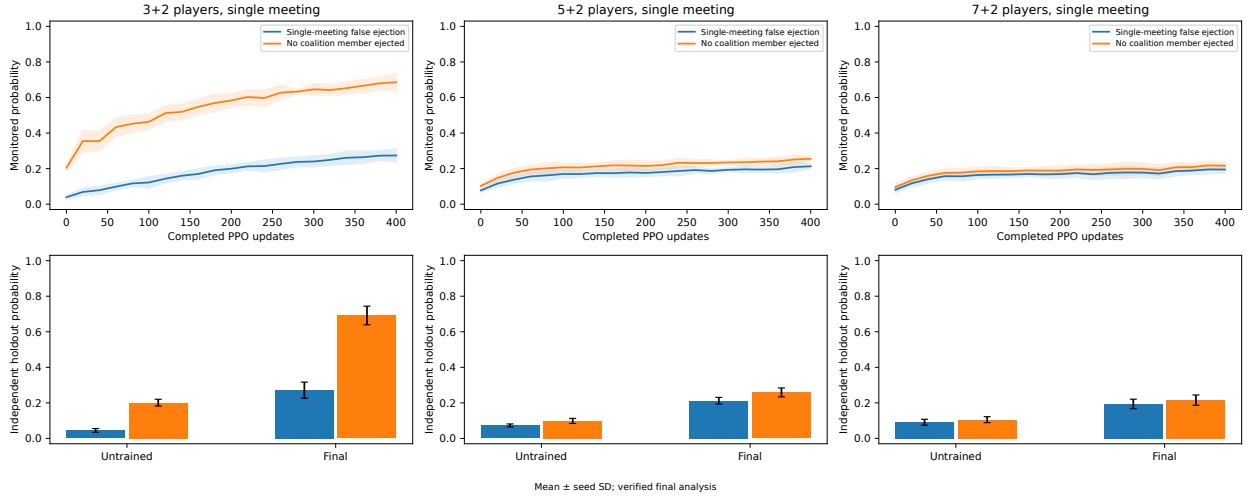


Figure 1. Meeting-only learning at all three populations. Curves use monitored evaluations; initial and final points use an independent holdout stream. Lines and error bars describe ten-seed variability, not independent episode replicates.

False ejection rises from 0.045 to 0.271 at 3+2, 0.073 to 0.2125 at 5+2, and 0.091 to 0.194 at 7+2. All three specified final-minus-initial false-ejection contrasts have Holm-adjusted $p = 0.0117$ within the six-test F1 family. Aligned voting and creator survival also increase descriptively, while the directly false-claim fraction falls at all sizes. These joint changes show that harmful coordination need not involve more explicitly false statements; they do not isolate communication as a causal driver.

Table 2. Independent holdout outcomes, mean $\pm$ sample SD across ten seeds. Aligned votes exclude shared abstention; creator survival is incident-gated. False claims count the FALSE label, separately from contradictions and no-information.

Players	Outcome	Untrained	Final
3+2	False ejection	0.045 ± 0.011	0.271 ± 0.046
3+2	Aligned votes	0.126 ± 0.023	0.613 ± 0.062
3+2	False claims	0.429 ± 0.017	0.328 ± 0.074
3+2	Creator survives	0.457 ± 0.024	0.789 ± 0.036
5+2	False ejection	0.073 ± 0.008	0.212 ± 0.018
5+2	Aligned votes	0.108 ± 0.023	0.316 ± 0.058
5+2	False claims	0.423 ± 0.012	0.349 ± 0.046
5+2	Creator survives	0.310 ± 0.020	0.438 ± 0.023
7+2	False ejection	0.091 ± 0.017	0.194 ± 0.027
7+2	Aligned votes	0.092 ± 0.016	0.244 ± 0.041
7+2	False claims	0.420 ± 0.011	0.385 ± 0.048
7+2	Creator survives	0.272 ± 0.018	0.366 ± 0.022

6.2 F2: testimony rules against scripted coalitions

Scripted coalition and crew-rule evaluation (5+2 players, single meeting)

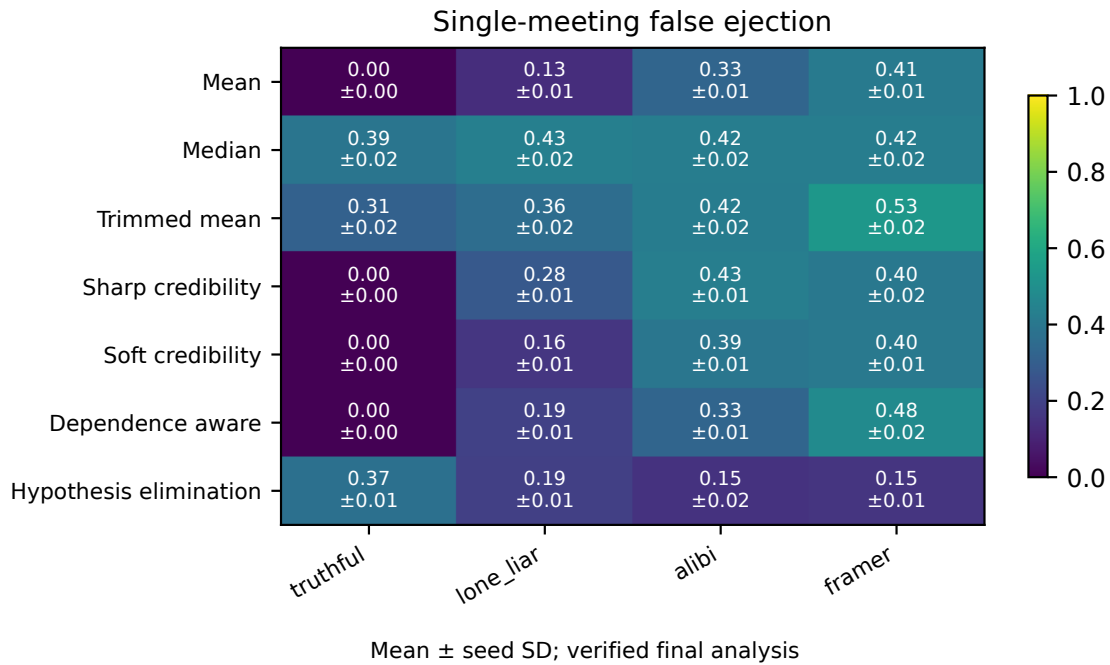


Figure 2. Seven rules against scripted testimony at 5+2, using ten replicates and 1,000 episodes per cell. False-ejection and creator-ejection outcomes answer different questions; the complete three-population values accompany the analysis.

Table 3. False-ejection probability. Entries are ten-replicate means; complete sample SDs and seed identities are in the accompanying machine-readable tables. Truthful concerns supported claim contents, while coalition voting and information selection remain adversarial.

Players / script	Mean	Median	Trim.	Soft	Sharp	Dep.	Hyp.
3+2 Truthful	0.000	0.137	0.137	0.000	0.000	0.000	0.245
3+2 Lone	0.328	0.245	0.245	0.178	0.204	0.210	0.412
3+2 Alibi	0.512	0.439	0.439	0.524	0.504	0.451	0.482
3+2 Framer	0.542	0.532	0.532	0.535	0.519	0.535	0.352
5+2 Truthful	0.000	0.389	0.312	0.000	0.000	0.000	0.370
5+2 Lone	0.131	0.430	0.359	0.159	0.278	0.194	0.191
5+2 Alibi	0.327	0.422	0.419	0.392	0.428	0.326	0.147
5+2 Framer	0.411	0.423	0.531	0.403	0.400	0.479	0.154
7+2 Truthful	0.000	0.504	0.380	0.000	0.000	0.000	0.393
7+2 Lone	0.100	0.553	0.417	0.151	0.300	0.178	0.070
7+2 Alibi	0.212	0.557	0.486	0.378	0.451	0.340	0.065
7+2 Framer	0.260	0.522	0.535	0.255	0.314	0.355	0.088

The repaired hypothesis rule has a substantial truthful-testimony cost: false ejection is 0.2447, 0.3700, and 0.3926 at 3+2, 5+2, and 7+2, while mean and soft credibility have zero observed errors in these cells. At 5+2, hypothesis scoring improves on the mean under alibi (0.1469 versus 0.3273) and framer (0.1542 versus 0.4106), but worsens lone-liar error (0.1907 versus 0.1311). Median and trimming can discard useful sparse testimony; their robustness to scalar contamination does not guarantee accurate votes here.

The F2 condition and rule families contain 63 and 36 contrasts. Their adjusted tests do not reject at 0.05 under the retained ten-seed resolution. Large descriptive differences should therefore be read with their reported between-replicate variability, not promoted into unreported familywise conclusions.

A bounded replay examined ten truthful-testimony false ejections from one 5+2 seed. The

Table 4. Incident-creator ejection probability. Entries are ten-replicate means; complete sample SDs and seed identities are in the accompanying machine-readable tables. Truthful concerns supported claim contents, while coalition voting and information selection remain adversarial.

Players / script	Mean	Median	Trim.	Soft	Sharp	Dep.	Hyp.
3+2 Truthful	0.052	0.036	0.036	0.052	0.052	0.052	0.019
3+2 Lone	0.178	0.165	0.165	0.171	0.155	0.142	0.158
3+2 Alibi	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3+2 Framer	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5+2 Truthful	0.901	0.415	0.521	0.901	0.901	0.901	0.475
5+2 Lone	0.661	0.358	0.461	0.563	0.407	0.490	0.555
5+2 Alibi	0.354	0.351	0.310	0.221	0.213	0.362	0.360
5+2 Framer	0.380	0.353	0.237	0.409	0.410	0.328	0.386
7+2 Truthful	0.930	0.314	0.492	0.930	0.930	0.930	0.495
7+2 Lone	0.779	0.279	0.453	0.657	0.485	0.606	0.834
7+2 Alibi	0.560	0.277	0.344	0.332	0.298	0.457	0.604
7+2 Framer	0.590	0.279	0.299	0.612	0.560	0.532	0.592

true coalition retained positive weight in all 40 honest pre-vote views, and the actual timelines satisfied the emitted constraints. Equal membership/creator scores could select an innocent through deterministic index tie breaking; in another case every honest voter skipped and coalition ballots alone decided the ejection. This establishes concrete failure routes, not their prevalence across the entire F2 distribution. Hard feasibility does not guarantee calibrated scores or accurate decisions.

6.2.1 Credibility weights and score interventions

We conducted a separate mechanism replication using the repaired frozen implementation: ten evaluation replicates with 500 meetings each for every combination of three initial crew sizes, four scripted conditions, three aggregation rules, and three variants (540,000 meetings). Tables 5 and 6 report means and sample SDs across replicates. These new measurements supersede the former single-seed mechanism diagnostic; they use a separate matched-evidence stream from the full F2 rule sweep above. Within each size, condition, and replicate, all nine rule/variant cells share the same exogenous evidence and episode identities. The interventions govern crew responses and votes, so subsequent public responses may differ; they do not hold the full transcript fixed.

“Coalition caught” is the fraction of honest voter decisions that privately refute at least one coalition claim. “Top pick differs” compares mean and soft credibility on the same soft-baseline record, restricted to eligible candidates. Credibility scores include refutation and self-contradiction penalties as well as pairwise compatibility. One crew member is removed by the incident before testimony: initial 3+2, 5+2, and 7+2 games have 2+2, 4+2, and 6+2 reporters. Appendix A counts actual reporters, so 3+2 is outside its strict-minority condition $m < n$.

Under truthful testimony, all speakers receive the same credibility score and the normalized weights are uniform; no same-record mean-versus-soft top-pick differences were observed. All truthful cells have zero observed false ejections, but this does not imply reliable identification at every size. In truthful 3+2 baseline meetings, the no-ejection rate is 0.9496 ± 0.0094 and the creator-ejection rate is 0.0504 ± 0.0094 . At truthful 5+2 and 7+2 every meeting ejects a coalition member, and creator-ejection rates are 0.9030 ± 0.0116 and 0.9204 ± 0.0113 ; remaining ejections select the partner.

For every attacked condition, the coalition’s average normalized weight is below its uniform share. In the coordinated alibi, the averages are 0.2630, 0.1425, and 0.0976 at initial 3+2, 5+2, and 7+2. Below-uniform averages alone do not exclude individual episodes of weight capture. A further audit of all 240,000 retained honest voter records found no coalition weight above its uniform share at numerical tolerance 10^{-12} . This is a descriptive result for these scripted records, not a general immunity claim. All speakers supplied informative claims in this experiment.

We tested two variants of each aggregation rule. “Alibi off” removes testimony-based alibi credit, including the positive WAS_WITH subject credit controlled by the same coefficient. “Catch outside” places the caught-lying suspicion penalty in the voter’s own evidence instead

Table 5. New pre-ejection soft-credibility diagnostic (mean $\pm$ SD across ten replicates of 500 meetings each). C/H denote coalition/honest speakers. Coalition weight is total normalized speaker weight; parentheses give its uniform share. All honest voters are included, including a voter subsequently ejected.

Game, condition	Credibility C	Credibility H	Coalition weight (uniform)	Coalition caught	Top pick differs
3+2, Truthful	3.000 $\pm$ 0.000	3.000 $\pm$ 0.000	0.5000 $\pm$ 0.0000 (0.5000)	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
3+2, Lone liar	1.581 $\pm$ 0.037	2.982 $\pm$ 0.006	0.4225 $\pm$ 0.0022 (0.5000)	0.4649 $\pm$ 0.0129	0.1264 $\pm$ 0.0076
3+2, Alibi	0.164 $\pm$ 0.075	2.960 $\pm$ 0.008	0.2630 $\pm$ 0.0059 (0.5000)	0.5583 $\pm$ 0.0147	0.1674 $\pm$ 0.0161
3+2, Framer	-1.547 $\pm$ 0.052	2.593 $\pm$ 0.040	0.1479 $\pm$ 0.0041 (0.5000)	0.7787 $\pm$ 0.0080	0.0029 $\pm$ 0.0014
5+2, Truthful	5.000 $\pm$ 0.000	5.000 $\pm$ 0.000	0.3333 $\pm$ 0.0000 (0.3333)	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
5+2, Lone liar	2.836 $\pm$ 0.054	4.987 $\pm$ 0.003	0.2606 $\pm$ 0.0017 (0.3333)	0.5353 $\pm$ 0.0138	0.0319 $\pm$ 0.0054
5+2, Alibi	0.778 $\pm$ 0.078	4.970 $\pm$ 0.004	0.1425 $\pm$ 0.0030 (0.3333)	0.6317 $\pm$ 0.0093	0.0769 $\pm$ 0.0085
5+2, Framer	-0.335 $\pm$ 0.081	4.476 $\pm$ 0.025	0.1062 $\pm$ 0.0035 (0.3333)	0.6921 $\pm$ 0.0125	0.0085 $\pm$ 0.0021
7+2, Truthful	7.000 $\pm$ 0.000	7.000 $\pm$ 0.000	0.2500 $\pm$ 0.0000 (0.2500)	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
7+2, Lone liar	4.068 $\pm$ 0.036	6.989 $\pm$ 0.002	0.1863 $\pm$ 0.0010 (0.2500)	0.5819 $\pm$ 0.0064	0.0174 $\pm$ 0.0036
7+2, Alibi	1.520 $\pm$ 0.124	6.973 $\pm$ 0.003	0.0976 $\pm$ 0.0027 (0.2500)	0.6535 $\pm$ 0.0104	0.0672 $\pm$ 0.0069
7+2, Framer	0.361 $\pm$ 0.057	6.402 $\pm$ 0.013	0.0630 $\pm$ 0.0018 (0.2500)	0.6830 $\pm$ 0.0080	0.0086 $\pm$ 0.0032

of within the speaker’s aggregated row. In the baseline credibility implementation, down-weighting a caught liar also reduces the self-incriminating penalty contained in that row.

Table 6. New false-ejection rates across all scripted conditions, rules and variants (mean $\pm$ SD across ten replicates of 500 meetings per cell). Soft and sharp denote credibility rules. Evidence is matched within each size/condition/replicate; responses and votes are regenerated. Differences are descriptive.

Game, condition	Rule	Baseline	Alibi off	Catch outside
3+2, Truthful	Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
3+2, Truthful	Soft	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
3+2, Truthful	Sharp	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
3+2, Lone liar	Mean	0.3290 $\pm$ 0.0179	0.3662 $\pm$ 0.0184	0.3290 $\pm$ 0.0179
3+2, Lone liar	Soft	0.1734 $\pm$ 0.0205	0.2106 $\pm$ 0.0237	0.3274 $\pm$ 0.0178
3+2, Lone liar	Sharp	0.2010 $\pm$ 0.0230	0.2188 $\pm$ 0.0277	0.3230 $\pm$ 0.0175
3+2, Alibi	Mean	0.5176 $\pm$ 0.0249	0.4030 $\pm$ 0.0219	0.5180 $\pm$ 0.0250
3+2, Alibi	Soft	0.5288 $\pm$ 0.0238	0.3728 $\pm$ 0.0193	0.5046 $\pm$ 0.0223
3+2, Alibi	Sharp	0.5144 $\pm$ 0.0252	0.4406 $\pm$ 0.0222	0.4528 $\pm$ 0.0201
3+2, Framer	Mean	0.5354 $\pm$ 0.0195	0.5370 $\pm$ 0.0200	0.5354 $\pm$ 0.0195
3+2, Framer	Soft	0.5294 $\pm$ 0.0200	0.5306 $\pm$ 0.0198	0.5294 $\pm$ 0.0200
3+2, Framer	Sharp	0.5124 $\pm$ 0.0202	0.5172 $\pm$ 0.0209	0.5110 $\pm$ 0.0204
5+2, Truthful	Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
5+2, Truthful	Soft	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
5+2, Truthful	Sharp	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
5+2, Lone liar	Mean	0.1326 $\pm$ 0.0181	0.1876 $\pm$ 0.0171	0.1326 $\pm$ 0.0181
5+2, Lone liar	Soft	0.1598 $\pm$ 0.0184	0.2064 $\pm$ 0.0199	0.1316 $\pm$ 0.0189
5+2, Lone liar	Sharp	0.2792 $\pm$ 0.0187	0.3244 $\pm$ 0.0154	0.1316 $\pm$ 0.0189
5+2, Alibi	Mean	0.3122 $\pm$ 0.0198	0.2312 $\pm$ 0.0224	0.3122 $\pm$ 0.0198
5+2, Alibi	Soft	0.3802 $\pm$ 0.0242	0.2310 $\pm$ 0.0215	0.2890 $\pm$ 0.0183
5+2, Alibi	Sharp	0.4198 $\pm$ 0.0282	0.3414 $\pm$ 0.0191	0.2676 $\pm$ 0.0210
5+2, Framer	Mean	0.4060 $\pm$ 0.0248	0.5244 $\pm$ 0.0236	0.4060 $\pm$ 0.0248
5+2, Framer	Soft	0.4014 $\pm$ 0.0275	0.4890 $\pm$ 0.0259	0.4010 $\pm$ 0.0276
5+2, Framer	Sharp	0.4054 $\pm$ 0.0214	0.4706 $\pm$ 0.0229	0.3790 $\pm$ 0.0261
7+2, Truthful	Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
7+2, Truthful	Soft	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
7+2, Truthful	Sharp	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
7+2, Lone liar	Mean	0.0908 $\pm$ 0.0179	0.1416 $\pm$ 0.0139	0.0908 $\pm$ 0.0179
7+2, Lone liar	Soft	0.1438 $\pm$ 0.0231	0.1834 $\pm$ 0.0169	0.0902 $\pm$ 0.0179
7+2, Lone liar	Sharp	0.3012 $\pm$ 0.0237	0.3672 $\pm$ 0.0205	0.0902 $\pm$ 0.0179
7+2, Alibi	Mean	0.2080 $\pm$ 0.0190	0.1632 $\pm$ 0.0191	0.2080 $\pm$ 0.0190
7+2, Alibi	Soft	0.3666 $\pm$ 0.0250	0.2090 $\pm$ 0.0157	0.1916 $\pm$ 0.0193
7+2, Alibi	Sharp	0.4362 $\pm$ 0.0256	0.4012 $\pm$ 0.0254	0.1786 $\pm$ 0.0199
7+2, Framer	Mean	0.2654 $\pm$ 0.0176	0.4670 $\pm$ 0.0212	0.2654 $\pm$ 0.0176
7+2, Framer	Soft	0.2574 $\pm$ 0.0180	0.3906 $\pm$ 0.0212	0.2568 $\pm$ 0.0176
7+2, Framer	Sharp	0.3162 $\pm$ 0.0163	0.4192 $\pm$ 0.0231	0.2412 $\pm$ 0.0172

Mean catch-outside false-ejection outcomes differ from baseline only in two 3+2 alibi episodes (0.5176 versus 0.5180): replays locate a 2.22×10^{-16} change in one candidate score that resolves a floating-point argmax tie and changes the meeting outcome. Algebraic equality therefore does not imply bitwise equality of decisions.

In the alibi condition, soft credibility exceeds the baseline mean’s false-ejection rate by paired differences of 0.0112 ± 0.0080 , 0.0680 ± 0.0136 , and 0.1586 ± 0.0112 at 3+2, 5+2, and 7+2. Moving the caught penalty outside reduces soft alibi false ejections by 0.0242 ± 0.0066 , 0.0912 ± 0.0150 , and 0.1750 ± 0.0084 relative to its own baseline. Removing alibi credit also lowers alibi false ejections for every tested rule and size. These interventions support a role for alibi credit and catch-penalty placement in the observed behavior, while allowing responses and resulting testimony use to change.

The improvements are conditional on the attack and size. Removing alibi credit raises lone-liar and framer false ejections; for soft credibility under the framer the increases are 0.0876 ± 0.0143 at 5+2 and 0.1332 ± 0.0222 at 7+2. Moving the caught penalty outside raises the soft lone-liar rate at 3+2 by 0.1540 ± 0.0188 , although it reduces lone-liar false ejections at the larger sizes. Soft framer rates change by at most 0.0006 under catch-outside. Sharper weighting also has no uniform advantage: at 7+2 its baseline false-ejection rate exceeds the mean by 0.2104 ± 0.0139 for the lone liar, 0.2282 ± 0.0172 for alibi, and 0.0508 ± 0.0124 for the framer.

False-ejection reductions can reflect abstention or tied votes rather than more correct ejections. Under coordinated alibi at 3+2, every tested rule/variant has creator-ejection rate zero and coalition-favorable rate one; reducing false ejections only increases no-ejection outcomes. Under the lone liar at 5+2, soft catch-outside lowers false ejection from 0.1598 to 0.1316, but raises no-ejection from 0.0402 ± 0.0049 to 0.1100 ± 0.0105 and coalition-favorable outcomes from 0.2000 ± 0.0207 to 0.2416 ± 0.0257 . Thus these measurements distinguish testimony-based alibi support and catch-penalty placement from increased coalition weight, without establishing a unique causal explanation or an improvement under every objective.

6.3 F3: single-meeting crossplay depends on opponent and population

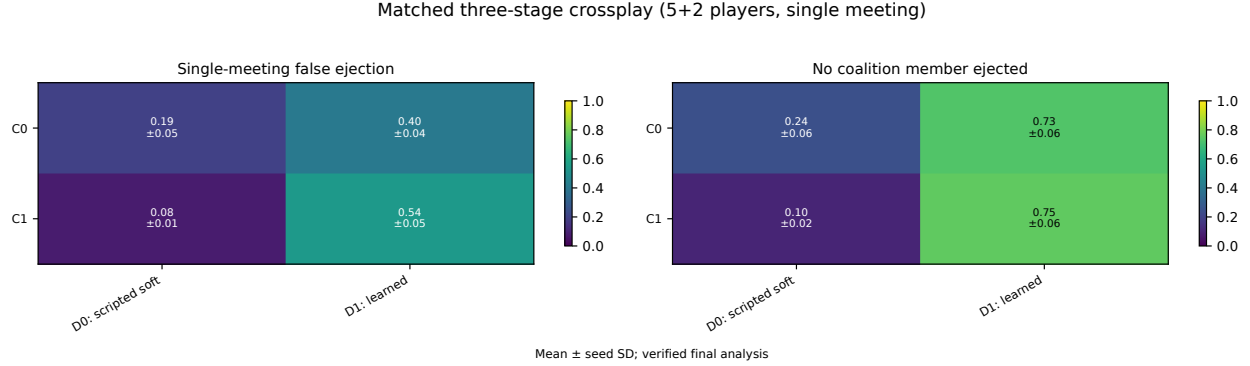


Figure 3. Spatial single-meeting crossplay at 5+2, including false ejection, incident opportunity, and creator outcomes. Each cell uses all ten seeds and 1,000 final games per seed.

Table 7. Unconditional per-game probabilities, mean $\pm$ sample SD across ten seeds. Creator ejection requires an incident. D0 is scripted; D1 and C1 are fresh policies trained against their frozen predecessor.

Players	Pair	False ejection	Incident	Creator ejected
3+2	C0–D0	0.188 $\pm$ 0.045	0.986 $\pm$ 0.005	0.128 $\pm$ 0.049
3+2	C0–D1	0.463 $\pm$ 0.073	0.087 $\pm$ 0.140	0.016 $\pm$ 0.030
3+2	C1–D1	0.415 $\pm$ 0.089	0.302 $\pm$ 0.286	0.025 $\pm$ 0.034
3+2	C1–D0	0.068 $\pm$ 0.034	0.949 $\pm$ 0.089	0.341 $\pm$ 0.101
5+2	C0–D0	0.194 $\pm$ 0.051	0.963 $\pm$ 0.027	0.566 $\pm$ 0.054
5+2	C0–D1	0.403 $\pm$ 0.040	0.254 $\pm$ 0.178	0.051 $\pm$ 0.035
5+2	C1–D1	0.538 $\pm$ 0.053	0.788 $\pm$ 0.132	0.092 $\pm$ 0.024
5+2	C1–D0	0.076 $\pm$ 0.014	0.984 $\pm$ 0.003	0.686 $\pm$ 0.036
7+2	C0–D0	0.211 $\pm$ 0.022	0.943 $\pm$ 0.030	0.584 $\pm$ 0.039
7+2	C0–D1	0.475 $\pm$ 0.037	0.321 $\pm$ 0.216	0.036 $\pm$ 0.026
7+2	C1–D1	0.587 $\pm$ 0.040	0.754 $\pm$ 0.117	0.061 $\pm$ 0.019
7+2	C1–D0	0.083 $\pm$ 0.020	0.970 $\pm$ 0.010	0.710 $\pm$ 0.032

At 5+2, false ejection rises from 0.1940 for C0–D0 to 0.4031 for C0–D1, then to 0.5385 for C1–D1. Against D0 the adapted coalition achieves only 0.0759. The specified defender-effect contrast is C0–D0 minus C0–D1; it is negative at all three sizes (−0.2752, −0.2091, −0.2638), with Holm-adjusted $p = 0.0176$. Thus the learned defender has more innocent ejections against the same frozen coalition.

Coalition adaptation against D1 has different signs across populations. At 3+2 the mean falls from 0.4635 to 0.4153, an effect of −0.0482 that does not reject; at 5+2 and 7+2 the increases are 0.1354 and 0.1124. Full paired intervals and adjusted tests are supplied alongside the tables. The data do not support a universal increase from coalition retraining.

C0 creates incidents in 0.986, 0.963, and 0.943 of games against D0, versus 0.087, 0.254,

and 0.321 against D1. Changing the defender also changes movement, opportunities, and testimony. Conditional error is therefore not an isolated reasoning comparison. Some 3+2 seeds have no incidents against D1: their incident-conditional rates remain undefined, while all unconditional summaries retain ten seeds. The formerly reported rare-incident account does not describe this corrected implementation; multiple repairs changed together, so their individual causal contributions are not identified.

6.4 F4: coalition victory and whole-game harm can diverge

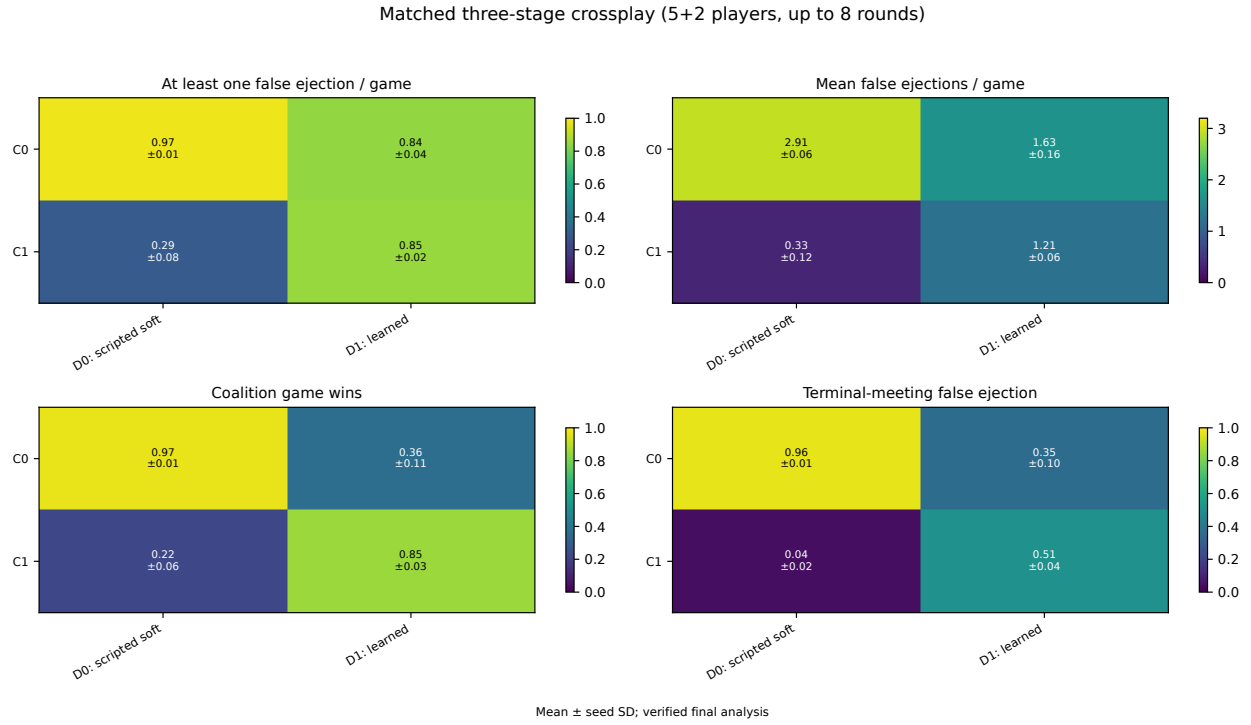


Figure 4. Multi-round crossplay at 5+2. Coalition victory, any innocent ejection, mean innocent-ejection count, and terminal outcomes are kept distinct. All ten seeds are included.

Table 8. Whole-game outcomes, mean $\pm$ seed SD. Any innocent ejection includes incident-free meetings; mean count is the number per game. It is not a terminal-meeting probability.

Players	Pair	Coalition wins	Any innocent ejected	Mean count
3+2	C0–D0	0.997 $\pm$ 0.004	0.562 $\pm$ 0.312	0.626 $\pm$ 0.410
3+2	C0–D1	0.282 $\pm$ 0.255	0.286 $\pm$ 0.197	0.324 $\pm$ 0.251
3+2	C1–D1	0.941 $\pm$ 0.058	0.521 $\pm$ 0.076	0.539 $\pm$ 0.093
3+2	C1–D0	0.988 $\pm$ 0.021	0.088 $\pm$ 0.014	0.088 $\pm$ 0.014
5+2	C0–D0	0.967 $\pm$ 0.012	0.974 $\pm$ 0.012	2.906 $\pm$ 0.059
5+2	C0–D1	0.360 $\pm$ 0.106	0.839 $\pm$ 0.043	1.634 $\pm$ 0.163
5+2	C1–D1	0.851 $\pm$ 0.026	0.847 $\pm$ 0.019	1.209 $\pm$ 0.060
5+2	C1–D0	0.218 $\pm$ 0.058	0.289 $\pm$ 0.084	0.332 $\pm$ 0.117
7+2	C0–D0	0.963 $\pm$ 0.007	0.978 $\pm$ 0.009	4.862 $\pm$ 0.042
7+2	C0–D1	0.304 $\pm$ 0.026	0.947 $\pm$ 0.006	2.996 $\pm$ 0.095
7+2	C1–D1	0.814 $\pm$ 0.016	0.947 $\pm$ 0.005	1.989 $\pm$ 0.053
7+2	C1–D0	0.063 $\pm$ 0.026	0.409 $\pm$ 0.054	0.558 $\pm$ 0.112

At 5+2, coalition victory drops from 0.9670 for C0–D0 to 0.3603 for C0–D1 and returns to 0.8514 for C1–D1. Yet any innocent ejection is 0.8392 and 0.8467 in the latter two pairings. The descriptive coalition-adaptation effects are +0.4911 for victory and only +0.0075 for any false ejection; mean innocent-ejection count falls by 0.4249 per game. At 7+2, victory rises by 0.5104 while mean count falls by 1.0066. Shorter or differently terminated games can therefore combine more coalition wins with fewer total innocent ejections. Reducing coalition wins therefore coexists with frequent crew harm; victory alone is an incomplete measure of defensive quality. Transfer again differs: C1–D0 wins only 0.2182 at 5+2 and 0.0630 at 7+2, but 0.9882 at 3+2.

None of the 27 F4 contrasts passes Holm correction at 0.05; the minimum adjusted value is 0.0527, imposed by the ten-seed exact-test resolution. The effect estimates remain relevant, but pointwise intervals do not override that familywise result. Whole-game harm records replace the old terminal-only interpretation.

6.5 Controls: information, defender budget, and reward design

Removing both aids gives single-meeting false ejection 0.2518 versus 0.1854 by default; this does not demonstrate that either aid is necessary for harm. None of the nine planned outcome-ablation contrasts rejects after Holm correction. The intervention does not remove shared parameters, rewards, prearranged behavior, or all possible role inference, and it tests only C0 rather than an entire response lineage.

Table 9. C0 information controls at 5+2, five seeds per condition. Hidden partner changes information and communication together; it does not prevent inference from observed events.

Condition	Single FE	Multi any-FE	Multi win
Default	0.185 ± 0.071	0.973 ± 0.011	0.967 ± 0.010
No channel	0.218 ± 0.071	0.983 ± 0.003	0.977 ± 0.004
Hidden partner	0.208 ± 0.059	0.973 ± 0.007	0.962 ± 0.008
Both removed	0.252 ± 0.038	0.970 ± 0.014	0.963 ± 0.014

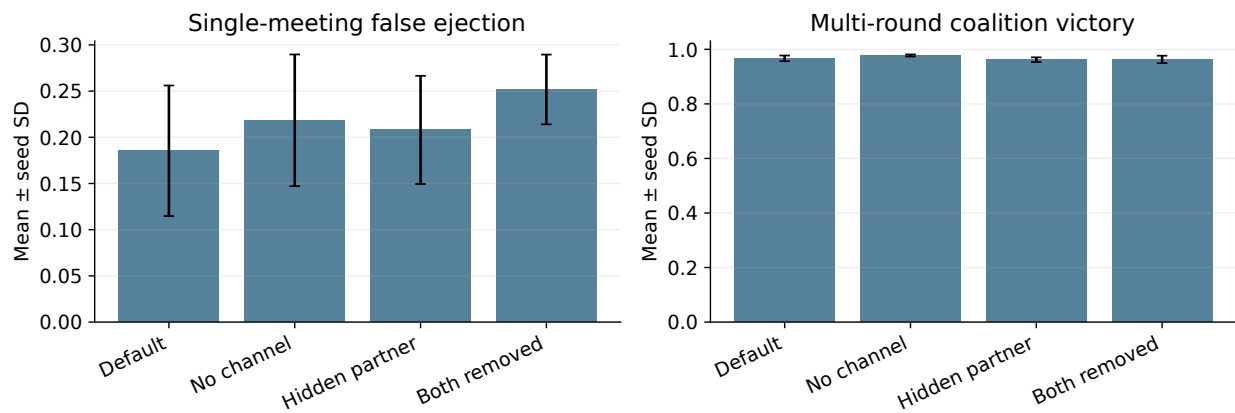


Figure 5. Information-control outcomes at 5+2, with sample SD across five seeds. Error bars are descriptive and do not establish equivalence.

Table 10. Defender-budget controls at 5+2. Every row, including the 400-update reference, uses exactly seeds 0–4. C0 and C1 keep 400 updates.

Outcome	Defender updates	C0–D1	C1–D1
Single FE	400	0.387 ± 0.040	0.525 ± 0.048
Single FE	800	0.439 ± 0.069	0.567 ± 0.055
Single FE	1600	0.382 ± 0.055	0.516 ± 0.039
Multi any-FE	400	0.840 ± 0.050	0.846 ± 0.012
Multi any-FE	800	0.745 ± 0.048	0.852 ± 0.034
Multi win	400	0.354 ± 0.115	0.854 ± 0.029
Multi win	800	0.165 ± 0.078	0.848 ± 0.029

In single-meeting play, C1–D1 false ejection is 0.5252 at 400 defender updates, 0.5672 at 800, and 0.5160 at 1,600 on the matched five seeds. None of the eight budget contrasts rejects after Holm correction. These finite budgets do not establish an optimal defender response, and the five-seed minimum raw p-value is already above 0.05.

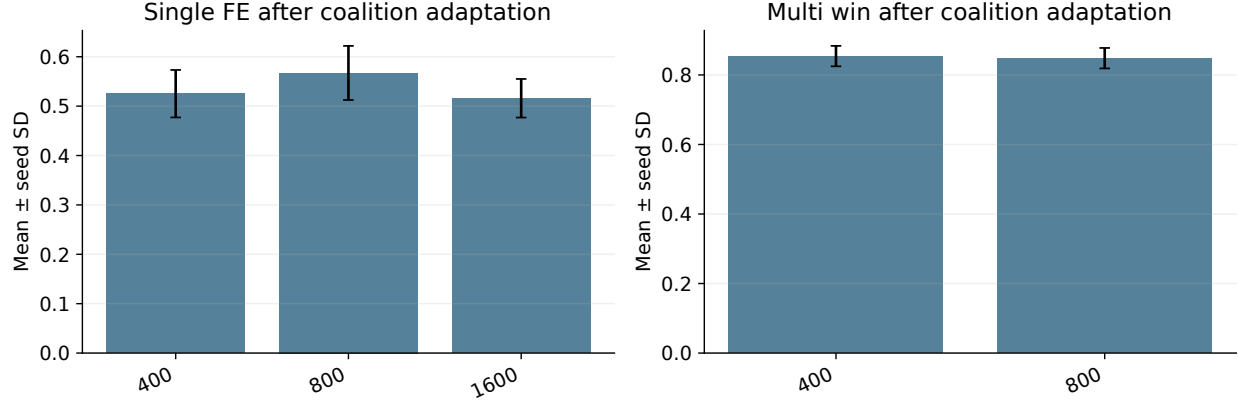


Figure 6. Matched-five-seed budget comparisons after coalition adaptation. The reference is recomputed from those five seeds, not the ten-seed headline estimate.

Table 11. Reward controls, means $\pm$ SD across the same five seeds and matched horizons: four generations for single-meeting, two for multi-round. C-stage and D-stage means omit C0; matrix gain averages $M[g,g]$ minus $M[g-1,g]$.

Outcome	Reward	C stages	D stages	Matrix gain
Single FE	Default	0.551 ± 0.040	0.267 ± 0.049	0.287 ± 0.075
Single FE	Balanced	0.548 ± 0.015	0.380 ± 0.018	0.164 ± 0.040
Single FE	Meeting	0.570 ± 0.051	0.347 ± 0.006	0.220 ± 0.049
Multi any-FE	Survival	0.854 ± 0.020	0.757 ± 0.060	0.101 ± 0.055
Multi any-FE	Framing count	0.888 ± 0.020	0.800 ± 0.029	0.090 ± 0.032
Multi win	Survival	0.810 ± 0.043	0.374 ± 0.076	0.440 ± 0.095
Multi win	Framing count	0.883 ± 0.014	0.554 ± 0.051	0.333 ± 0.060

None of the 18 specified reward-control contrasts rejects after Holm correction. The variants can alter outcome levels and response gains descriptively; the evidence does not show that reward design is irrelevant. Five seeds provide limited resolution, and the tested objectives do not exhaust alternative incentives or learning algorithms.

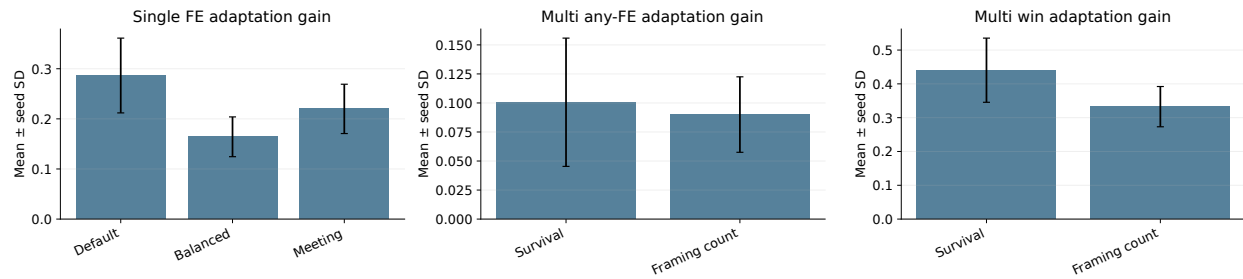


Figure 7. Reward-control matrix adaptation gains on matched seed cohorts and horizons. Panels report different outcomes, not a common reward scale.

6.6 F5: repeated gains over a finite response sequence

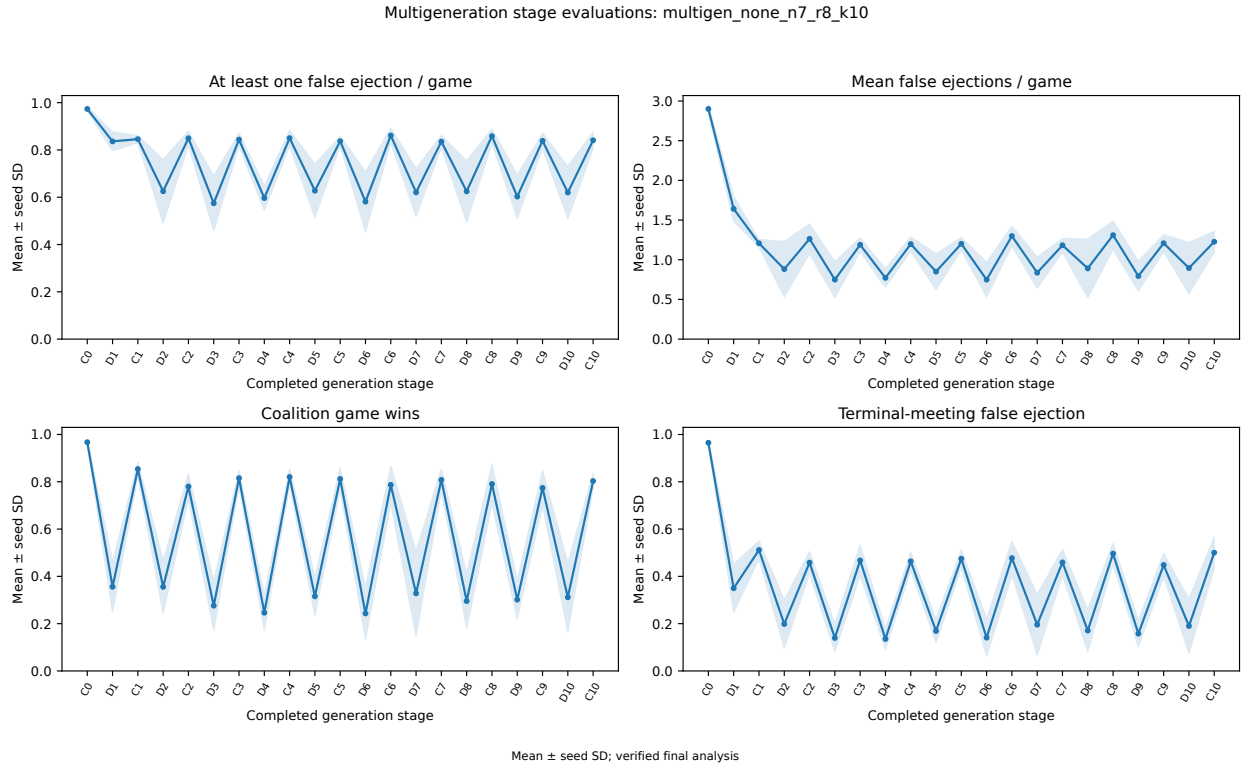


Figure 8. The 5+2 multi-round sequence through ten generations, ten seeds. C and D stages are independently initialized response policies; monitored and holdout quantities remain separate.

Table 12. Finite-horizon response summaries. Single-meeting uses false ejection; multi-round uses coalition victory. Stage means omit C0. Matrix gains use final crossplay, not differences between monitored curves.

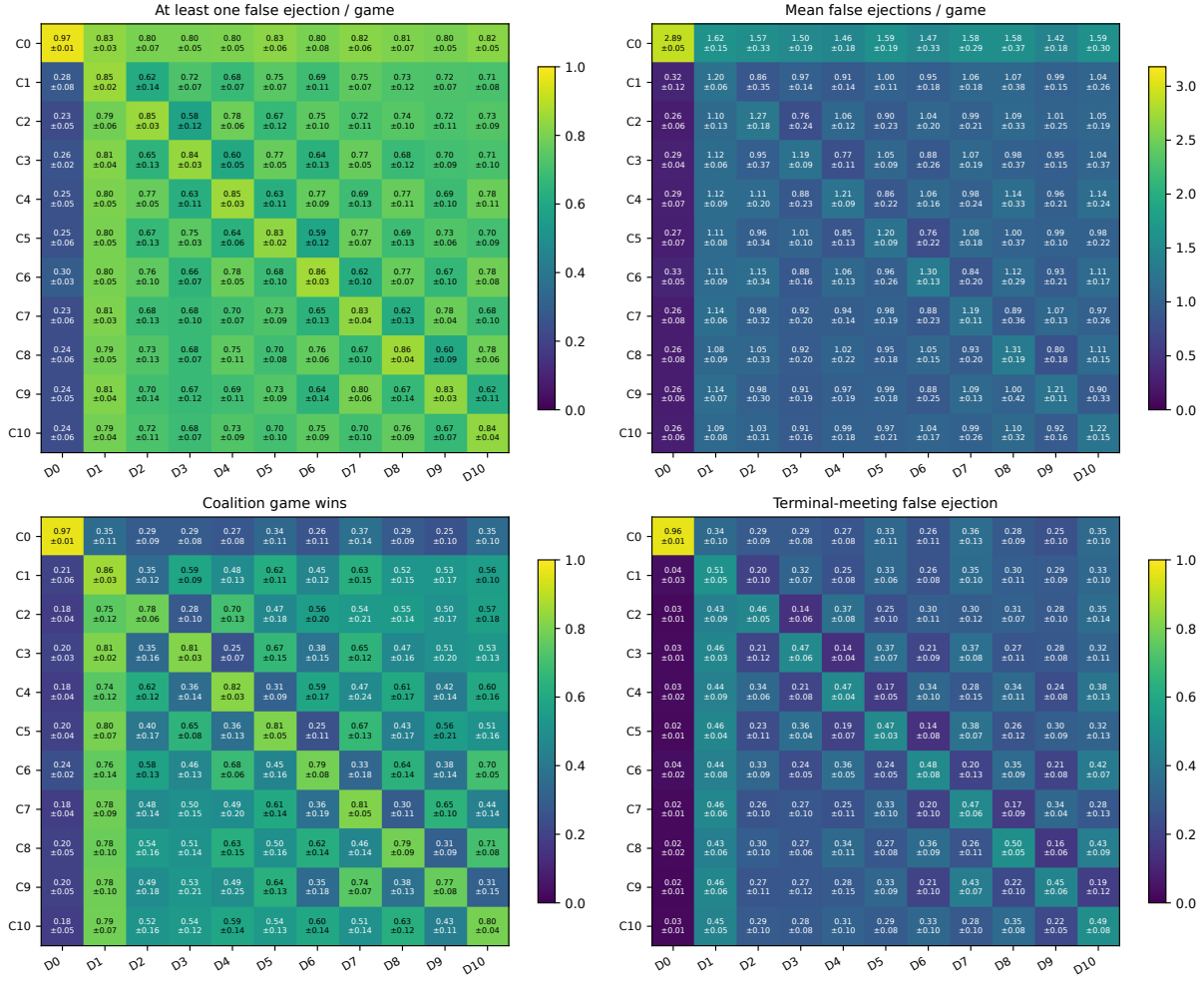
Setting	Seeds	C stages	D stages	Matrix gain
3+2 multi	10	0.918 ± 0.021	0.341 ± 0.059	0.576 ± 0.050
5+2 single	10	0.557 ± 0.016	0.257 ± 0.023	0.300 ± 0.032
5+2 multi	10	0.804 ± 0.025	0.303 ± 0.035	0.501 ± 0.050
7+2 multi	5	0.752 ± 0.048	0.330 ± 0.044	0.418 ± 0.088

At 5+2, coalition victory is 0.9658 for C0–D0, 0.3524 for C0–D1, and 0.8562 for C1–D1. By generation ten it is 0.3068 for C9–D10 and 0.7968 for C10–D10, while C10–D0 achieves only 0.1804. The coalition repeatedly improves against the newest defender, with limited transfer to the scripted policy. The ten-seed loops’ specified matrix-gain tests pass Holm correction in the 25-test dynamics family ($p = 0.0488$); the five-seed largest-population loop does not. None of the specified late-minus-early or parity tests rejects after correction.

Policy distances and outcome reversals do not identify an attracting cycle. The policies are independently initialized at each stage, the opponent changes, and outcome probabilities are estimated with finite precision. A small observed gap can also reflect a limited policy pool. These data describe repeated adaptive vulnerability within the measured horizon, without proving convergence, nonconvergence, optimality, or permanent attacker advantage.

Two terminal coordination cells in the 7+2 loop are undefined because all 500 games in each cell lost a coalition voter before the last meeting. Complete meeting histories verify the zero denominator. No null was replaced by zero, no seed was dropped, and all primary crossplay outcomes remain complete. Terminal coordination should not be confused with coordination over every meeting of a game.

Cross-generation evaluation matrix (5+2 players, up to 8 rounds)



Mean $\pm$ seed SD; verified final analysis

Figure 9. Full 5+2 multi-round crossplay, 500 episodes per cell and ten seeds. Comparing each diagonal entry with the entry immediately above measures adaptation against a fixed defender.

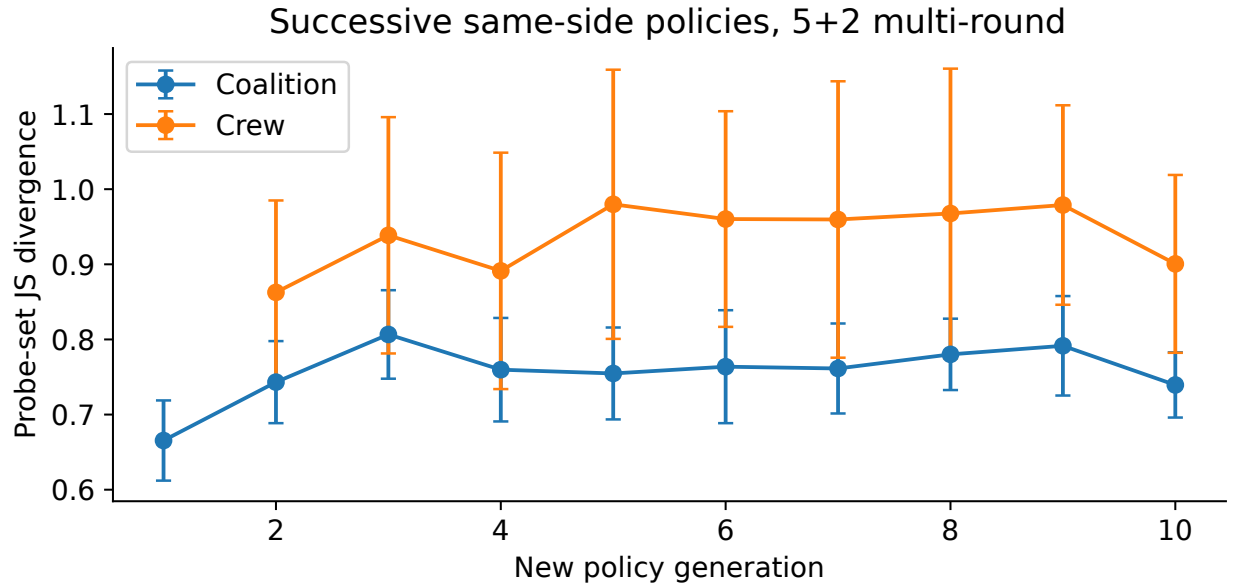


Figure 10. Jensen–Shannon distances between successive same-side policies on fixed probe observations, mean $\pm$ ten-seed SD. Learned defender comparisons begin at D2 versus D1; D0 is scripted. Distances reflect fresh initialization and training and are not distances along a single continuous optimization trajectory.

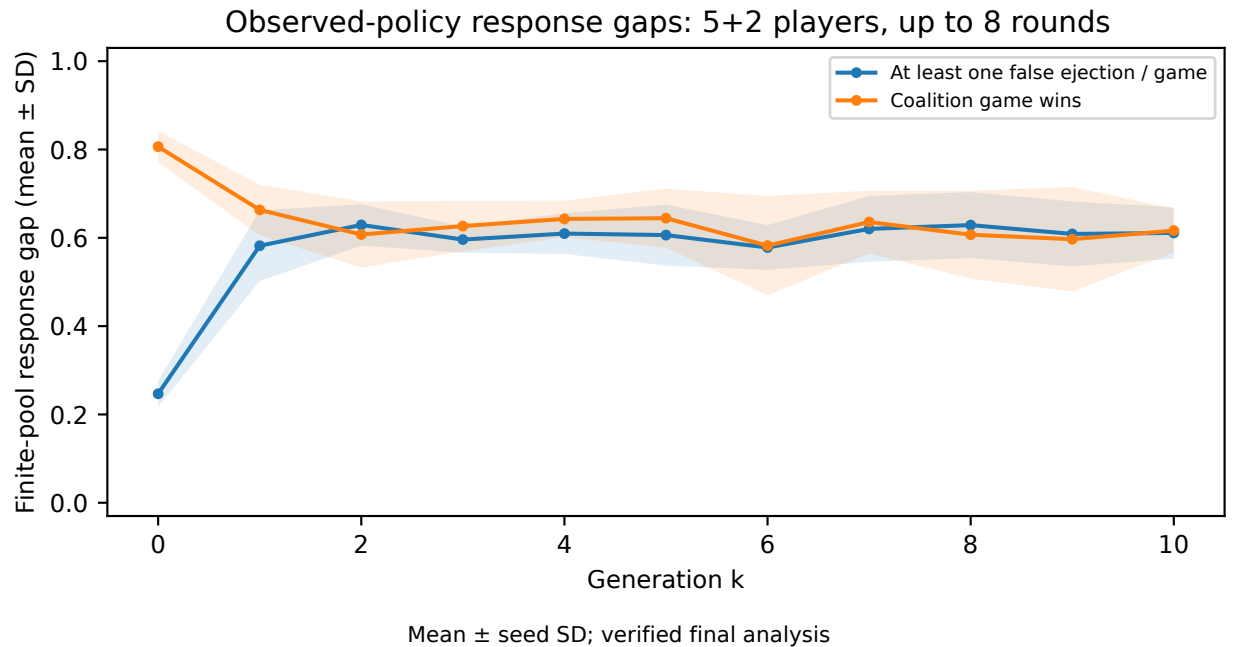


Figure 11. Retrospective response gaps in the finite observed policy pool. Conservative simultaneous cell bands account for Monte Carlo error and selection of extrema; later-generation policies are included in earlier-generation pools.

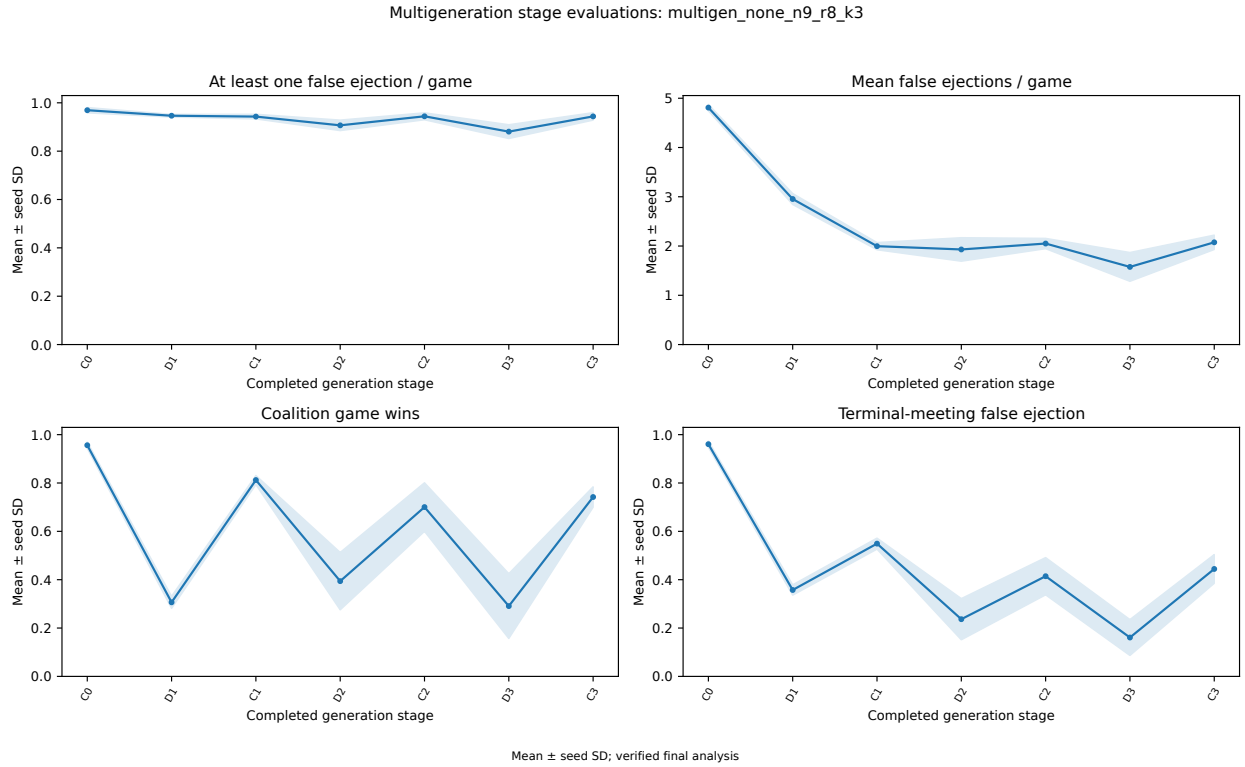


Figure 12. Largest-population sequence, 7+2 players and three generations across five seeds. This shorter horizon and coarser statistical resolution limit comparison with ten-generation results.

6.7 F6: an analytical model of corroboration capture under correlated testimony

The model isolates the one failure mode that the framing of “credibility collapse” usually refers to: manufactured agreement earning a coalition more than its share of a trust rule’s weight. We use the term in that sense only; Section 6.2’s corrected scripted diagnostics do not show average coalition-weight amplification in those conditions. The model replaces claims and votes by real-valued reports about a latent state θ . n honest agents report $x_i = \theta + \varepsilon_i$ with $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$ i.i.d.; m coalition agents report $y_j = \theta + b + \eta_j$ with equicorrelated Gaussian noise of marginal variance σ_c^2 , independent of honest noise, and pairwise correlation ρ ($\rho = 1$: identical, fully coordinated reports); $f = m/(n + m)$ is the coalition fraction. The corroboration count $c_i = \#\{k \neq i : |r_i - r_k| \leq h\}$ is the analogue of counting speakers whose claims are jointly satisfiable with one’s own, and the credibility-weighted estimate is $\hat{\theta} = \sum_i w_i r_i$ with $w_i \propto e^{\gamma c_i}$, gain $\gamma = 1/\tau$. Because differences of Gaussians are Gaussian, the pairwise agreement probabilities are explicit, $p_{HH} = \text{erf}(h/2\sigma)$, $p_{CC}(\rho) = \text{erf}(h/(2\sigma_c\sqrt{1-\rho}))$, $p_{CC}(1) = 1$, $p_{HC}(b) = \Phi(\frac{h-b}{s}) - \Phi(\frac{-h-b}{s})$ with $s^2 = \sigma^2 + \sigma_c^2$, and the expected corroboration counts are $\bar{c}_H = (n-1)p_{HH} + m p_{HC}$ and $\bar{c}_C = (m-1)p_{CC}(\rho) + n p_{HC}$. In mean field the coalition’s total weight is the logistic

$$W_C(\gamma, \rho, f) = \frac{f}{f + (1-f)e^{-\gamma\Delta c(\rho)}}, \quad \Delta c = \bar{c}_C - \bar{c}_H.$$

Appendix A separates exact Gaussian agreement probabilities from a deterministic mean-field surrogate obtained by applying softmax to expected counts. For positive gain, the surrogate gives the coalition more than its uniform share exactly when $\Delta c > 0$. For $b \neq 0$, this increases its squared bias; its MSE also contains honest and coalition variance. The Gaussian quantile $\sigma\Phi^{-1}(1/[2(1-f)])$ is an asymptotic reference for the median’s worst-case bias, so its intersection with $W_C|b|$ is a bias comparison, not a finite-sample error guarantee. With independent repetitions, in the long-window limit and on a correlation interval with $p_{CC} \geq p_{HC}$ throughout, a dependence strength $\lambda \geq \gamma(m-1)(N-1)/n$ makes the defended surrogate’s coalition weight non-increasing with correlation on that interval, where $N = n + m$. The realized softmax can have a different amplification boundary.

Numerically, with $n = 5$, $m = 2$, $\sigma_c = \sigma$, $h = 0.4\sigma$ (honest reporters agree with probability 0.22), $b = 4\sigma$ and $\gamma = 1$, credibility weighting is the best summing rule against *independent* liars (mean-squared error 0.87 against 1.45 for the mean and 0.57 for the median) and the worst rule against the *same* liars coordinating (1.84 at $\rho = 1$), while the median has similar rounded error at those endpoints (0.57): the surrogate has $\Delta c = -0.6$ at $\rho = 0$ and $+0.12$ at $\rho = 1$. The dependence-aware rule brings the coordinated case to 0.31, below the median, but pays for it at short windows and at low correlation (1.02 at $\rho = 0$), because the concentration statistic is estimated from few rounds. In the equal-variance, minority-coalition surrogate, independent biased reporters cannot amplify their weight: at $\rho = 0$, $\Delta c = (n-m)(p_{HC} - p_{HH}) \leq 0$. Unequal variances and finite-sample softmax fluctuations require separate analysis. The median retains a uniform bound in coalition bias whenever $m < n$; this worst-case guarantee is lost at one half, which does not mean every estimator

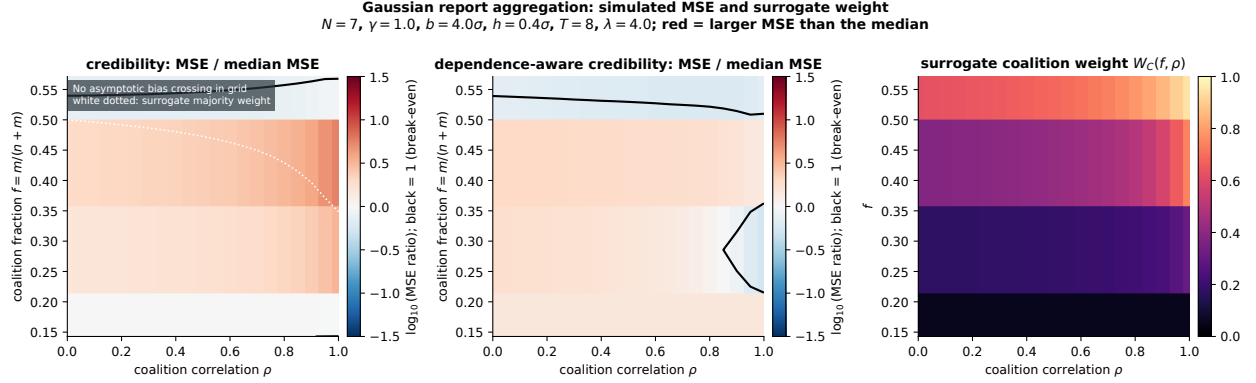


Figure 13. Phase diagram of the model ($N = 7$ reporters, $\gamma = 1$, $b = 4\sigma$, $h = 0.4\sigma$, $\sigma_c = \sigma$). Rows are discrete coalition sizes, $f = 1/7, 2/7, 3/7, 4/7$; contours interpolate the sampled grid. The row with $f \geq 1/2$ lies outside Appendix A’s minority-coalition guarantee. Colour: $\log_{10}$ of the ratio of the estimator’s error to the median’s; black contour: break-even; white reference contours, where present, mark the surrogate’s asymptotic bias-reference crossing and majority-weight boundary. The $N=7$ grid has no bias-reference crossing. These are analytical references, not exact boundaries for the realized estimator. Left: credibility weighting; middle: dependence-aware credibility (window of eight rounds, $\lambda = 4$); right: the mean-field coalition weight $W_C(f, \rho)$.

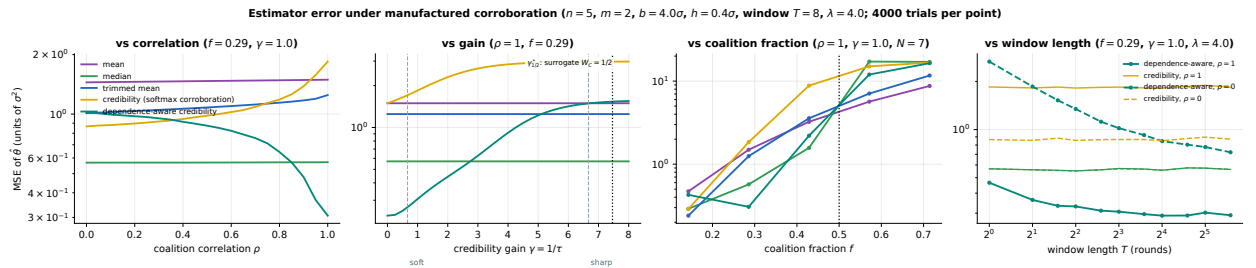


Figure 14. Estimator error (units of σ^2) against coalition correlation, credibility gain (the game’s soft and sharp gains marked), coalition fraction, and the window length of the dependence penalty; 5+2 defaults with $\sigma_c = \sigma$, 4,000 trials per point. The coalition-fraction sweep includes values outside the minority-coalition guarantee.

fails on every distribution.

Expected counts and expected weights can even predict different directions of amplification. For $n = 5$, $m = 2$, $\sigma = \sigma_c = 1$, $b = 20$, $h = 0.43$, $\rho = 1$, $\gamma = 1$, the surrogate gives $W_C = 0.295 > 2/7$, while 100,000 Monte Carlo trials give mean realized coalition weight 0.277, below the uniform share. The model supplies a mechanism and analytical references; the game’s realized weights must be checked directly.

Correspondence with the game. The scripted mechanism diagnostic directly measures lower average coalition weight than the uniform share in the tested alibi cells. This is compatible with the surrogate’s prediction when honest agreement is high, but does not validate a quantitative mapping from the Gaussian report model to symbolic testimony. The game also subtracts caught-lying penalties from credibility scores, so Table 5’s scores are not pure pairwise agreement counts. Neither the alibi-witness term nor the placement of the catch penalty is represented in the Gaussian model. The measured game weights, rather than the surrogate alone, establish the absence of average coalition-weight amplification in those cells. The initial 3+2 game has only 2+2 reporters after its incident victim is removed, so it lies outside the proposition’s strict-minority condition; the model’s n and m refer to actual reports, not initial players.

A sensor-fusion analogue. The rule was also applied to twelve sensors reporting a drifting scalar, with three compromised sensors (bias 3σ , $\rho = 1$, tolerance 0.4σ). Across 20 runs, credibility weighting assigns the coalition 0.254 ± 0.006 of total weight, only slightly above its uniform share of 0.25, compared with 0.099 ± 0.004 under independent compromised reports. Under coordination its RMSE is $0.99 \pm 0.03 \sigma$, versus 0.83 ± 0.02 for the mean and 0.58 ± 0.02 for the median. The dependence-aware variant reduces coalition weight to 0.044 and RMSE to $0.56 \pm 0.01 \sigma$. With independent compromised reports, the median remains best (0.58 ± 0.02), followed by credibility (0.60 ± 0.02), dependence-aware credibility (0.61 ± 0.02), and the mean (0.81 ± 0.01). These simulations illustrate a benefit of the dependence penalty under the chosen parameters; they do not establish a general repair guarantee for trust aggregation.

6.8 F7: dependence-aware defense changes distinct outcomes differently

Scripted coalition and crew-rule evaluation (5+2 players, single meeting)

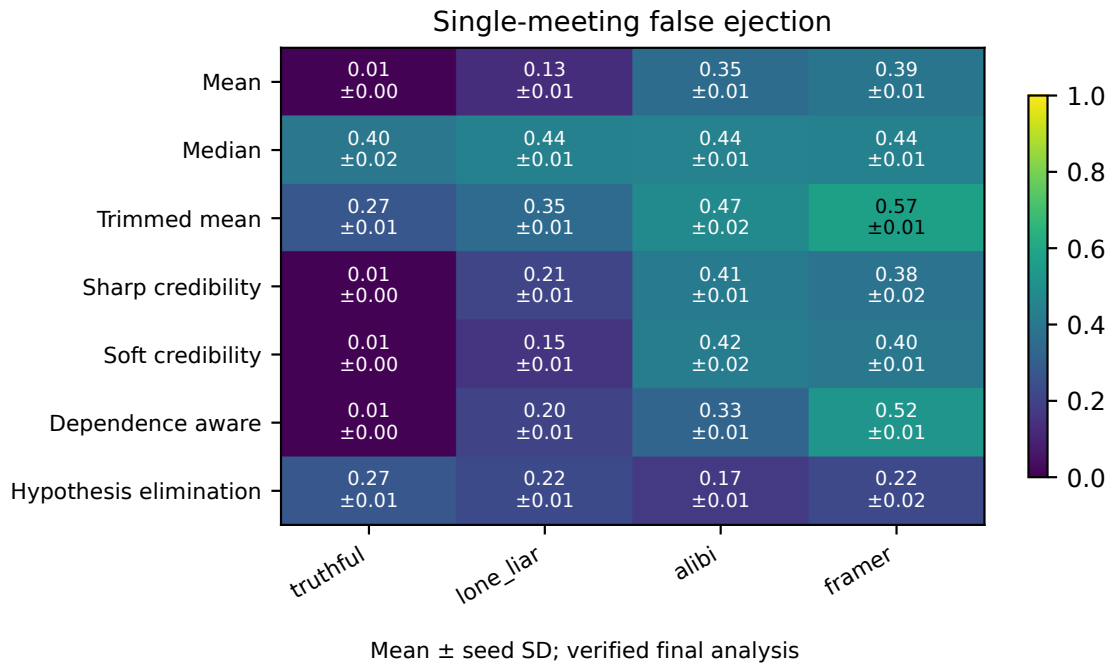


Figure 15. Spatial static-rule evaluations at 5+2, five evaluation replicates. This configuration has action-dependent evidence and differs from F2's exogenous meeting-only evidence.

Table 13. Spatial static false-ejection rates, mean $\pm$ SD across five seeds. C0–C3 are frozen checkpoints from the single-meeting loop; no rule is tuned by selecting its best test strength. The full seven-rule and strength sweep is included with the data.

Coalition	Mean	Soft	Dependence	Hypothesis
Truthful	0.007 ± 0.003	0.007 ± 0.003	0.007 ± 0.003	0.270 ± 0.010
Lone	0.132 ± 0.008	0.151 ± 0.011	0.204 ± 0.011	0.224 ± 0.014
Alibi	0.352 ± 0.013	0.421 ± 0.015	0.328 ± 0.012	0.168 ± 0.007
Framer	0.388 ± 0.011	0.399 ± 0.014	0.521 ± 0.014	0.222 ± 0.015
C0	0.145 ± 0.068	0.193 ± 0.067	0.185 ± 0.072	0.111 ± 0.061
C1	0.038 ± 0.014	0.074 ± 0.021	0.062 ± 0.020	0.057 ± 0.028
C2	0.058 ± 0.018	0.092 ± 0.021	0.086 ± 0.022	0.061 ± 0.021
C3	0.045 ± 0.016	0.080 ± 0.020	0.076 ± 0.023	0.050 ± 0.022

Table 14. Adaptive dependence crossplay. Rule discounts testimony; rule + tally also discounts votes. Values are mean $\pm$ SD across ten seeds at 5+2 and five at 3+2, with 1,000 episodes per cell.

Players	Trained against	Soft	Rule	Rule + tally
3+2	soft	0.210 ± 0.048	0.206 ± 0.038	0.413 ± 0.040
3+2	rule	0.194 ± 0.030	0.226 ± 0.036	0.439 ± 0.016
3+2	both	0.025 ± 0.006	0.039 ± 0.017	0.872 ± 0.010
5+2	soft	0.194 ± 0.055	0.182 ± 0.051	0.204 ± 0.052
5+2	rule	0.174 ± 0.031	0.170 ± 0.038	0.193 ± 0.034
5+2	both	0.169 ± 0.038	0.164 ± 0.042	0.190 ± 0.040

At 5+2, rule-specific retraining gives false ejection 0.1696 against the dependence rule, versus 0.1819 for the soft-trained coalition: effect -0.0123, pointwise 95% CI [-0.0516, 0.0374], Holm-adjusted $p = 1$. Against the combined rule/tally, retraining changes 0.2044 to 0.1896: effect -0.0148, CI [-0.0513, 0.0215], adjusted $p = 1$. These corrected comparisons do not reproduce a general defense-specific attack gain. At 3+2, however, the combined-defense rate rises descriptively from 0.4132 to 0.8722; with five seeds the adjusted result remains nonsignificant. Neither outcome establishes immunity.

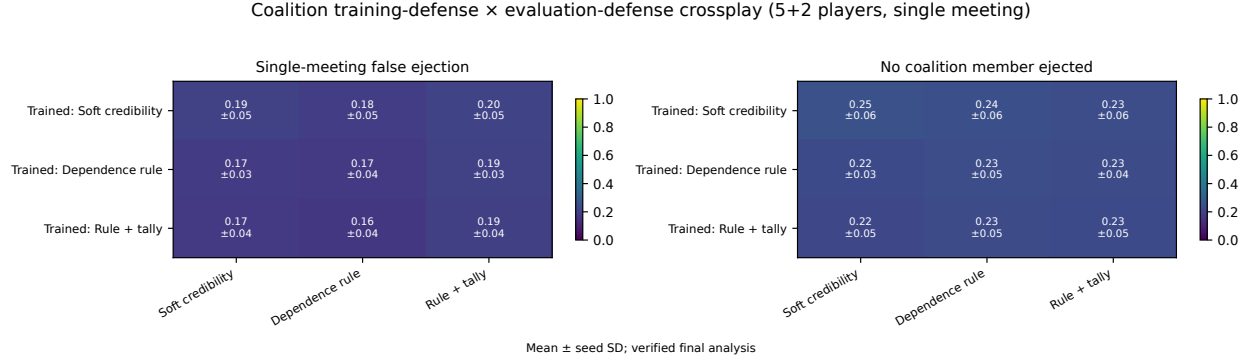


Figure 16. Dependence-specific coalition training and transfer at 5+2. Rows identify the training defense; columns identify the evaluation defense.

Table 15. Dependence-aware loop versus the undefended loop over the same four generations and ten seeds. Difference is defended minus reference; stages exclude C0. Means $\pm$ SD and adjusted tests retain the six-contrast family.

Outcome	Summary	Reference	Defended	Difference	Holm p
Any FE	C stages	0.847 $\pm$ 0.009	0.890 $\pm$ 0.007	+0.042	0.0117
Any FE	D stages	0.658 $\pm$ 0.051	0.763 $\pm$ 0.047	+0.104	0.0117
Any FE	Matrix gain	0.191 $\pm$ 0.051	0.129 $\pm$ 0.046	-0.062	0.0234
Coalition win	C stages	0.817 $\pm$ 0.025	0.789 $\pm$ 0.026	-0.029	0.0410
Coalition win	D stages	0.309 $\pm$ 0.065	0.348 $\pm$ 0.054	+0.040	0.1387
Coalition win	Matrix gain	0.509 $\pm$ 0.086	0.445 $\pm$ 0.070	-0.064	0.0469

The dependence-aware loop increases any innocent ejection by 0.042425 after C stages (point-wise 95% CI [0.034775, 0.049400]) and by 0.104375 after D stages ([0.073274, 0.135025]); both have Holm-adjusted $p = 0.0117$. At C stages it reduces coalition victory by 0.028775 ([-0.045625, -0.012050], adjusted $p = 0.0410$). Matrix adaptation gains also shrink, but this is not a guarantee of less harm or better stability. The same defense can reduce one success measure while increasing another adverse outcome.

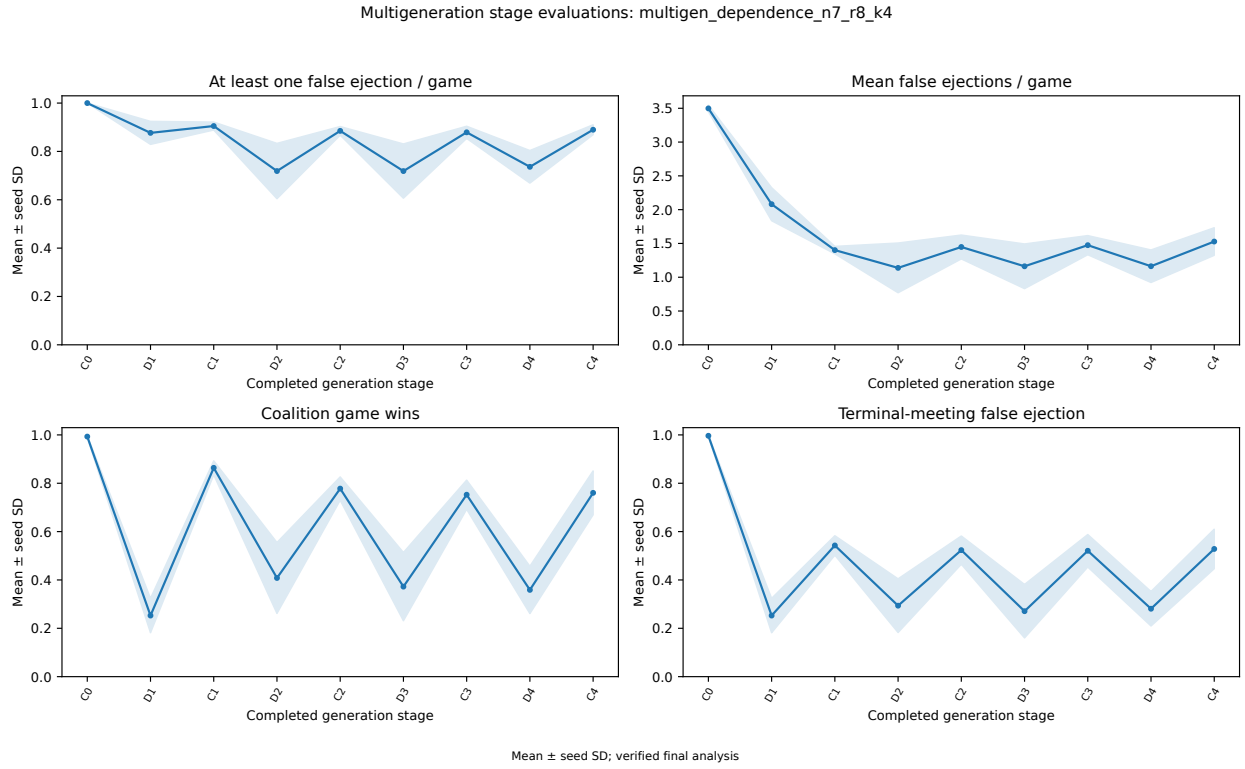


Figure 17. The defended four-generation 5+2 multi-round sequence. Table 15 compares it to the matching horizon and seeds of the undefended sequence.

6.9 F8: adaptive hypothesis attacks and ballot diagnostics

Table 16. Adaptive hypothesis crossplay, ten seeds at 5+2 and five at 7+2, 1,000 independent final episodes per cell. Training against a rule is distinct from transferring a coalition trained against another rule.

Players	Trained against	Mean	Soft	Hypothesis
5+2	mean	0.104 ± 0.037	0.148 ± 0.040	0.077 ± 0.031
5+2	soft	0.143 ± 0.053	0.194 ± 0.055	0.126 ± 0.058
5+2	hypothesis	0.096 ± 0.063	0.098 ± 0.068	0.422 ± 0.066
7+2	mean	0.139 ± 0.051	0.189 ± 0.056	0.124 ± 0.074
7+2	soft	0.136 ± 0.017	0.202 ± 0.013	0.110 ± 0.033
7+2	hypothesis	0.037 ± 0.007	0.030 ± 0.007	0.604 ± 0.018

At 5+2, targeted training increases hypothesis-defender false ejection from 0.1255 to 0.4217: paired effect 0.2962, pointwise 95% CI [0.2521, 0.3418], Holm-adjusted $p = 0.0273$. Its matched adaptive error also exceeds soft-trained versus soft by 0.2275 (adjusted $p = 0.0273$). Yet the hypothesis-trained coalition achieves only 0.0962 against mean and 0.0976 against soft credibility. This is a substantial tested adaptive vulnerability with opponent specialization, not a uniformly stronger attack.

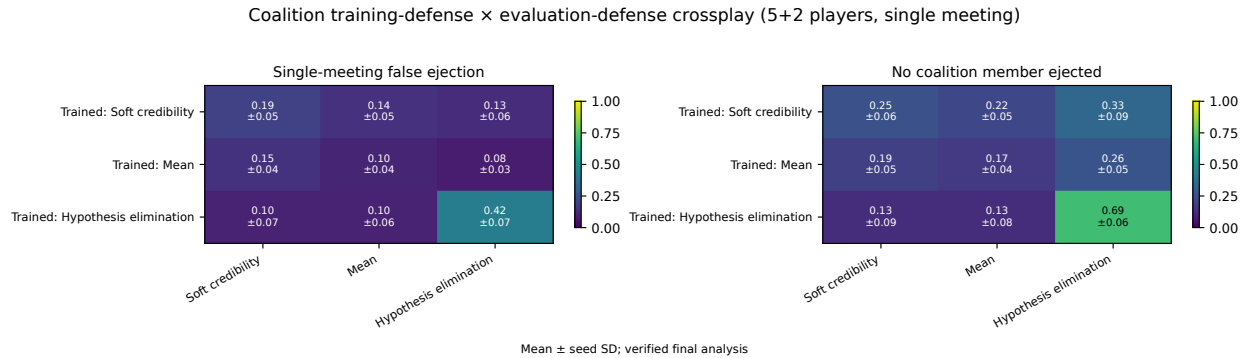


Figure 18. Adaptive hypothesis attack and transfer at 5+2. The matrix uses the main final-evaluation stream; the ballot supplement below uses an independent stream.

Table 17. Independent ballot supplement evaluated against the hypothesis defender. Entries are means of within-seed rates; full SDs and defined-seed counts accompany the data. Necessary: removing coalition ballots prevents an observed false ejection. Sufficient: coalition ballots alone select an innocent. Both condition on observed false ejections and hold votes fixed; they are different counterfactuals. Top score is the honest view’s leading membership score, not a calibrated probability.

Players	Attack	FE	Crew skip	Top score	Necessary	Sufficient	Crew-only FE	Seeds
5+2	mean	0.075	0.196	0.794	0.651	0.603	0.032	10
5+2	soft	0.117	0.271	0.746	0.684	0.662	0.040	10
5+2	hypothesis	0.424	0.661	0.507	0.973	0.978	0.014	10
7+2	mean	0.128	0.448	0.683	0.770	0.769	0.027	5
7+2	soft	0.101	0.406	0.713	0.819	0.823	0.022	5
7+2	hypothesis	0.605	0.899	0.410	1.000	1.000	0.000	5

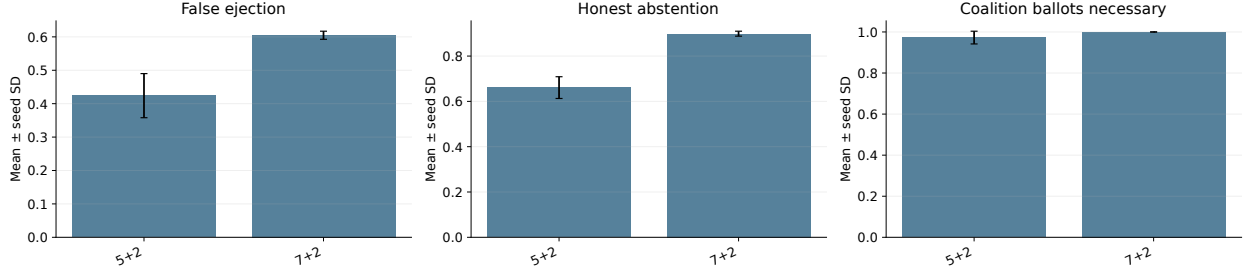


Figure 19. Hypothesis-trained attacks against the hypothesis rule in the independent ballot supplement. Means $\pm$ sample SD use ten seeds at 5+2 and five at 7+2. Ballot necessity is conditional on a false ejection.

The separate 135-cell, 67,500-episode diagnostic gives false ejection 0.4240 at 5+2 and 0.6048 at 7+2 for hypothesis-trained coalitions. Honest voters skip at rates 0.6608 and 0.8988. Removing coalition ballots prevents 97.26% and 100% of observed false ejections; coalition ballots alone are sufficient in 97.75% and 100%. Crew-only false ejection is 0.0142 and 0, respectively. Raw verification covers the full electorate, including the ejected voter, and recomputes tally and score-derived quantities.

These results support an abstention-related route to failure under recorded votes. They do not show that abstention is the only cause, that all score ties are uninformative, or that agents would retain their votes if the decision rule changed. Membership scores are feasibility summaries, not exact posterior probabilities; their numerical scale alone does not establish calibrated uncertainty.

7 Interpretation

The completed experiments distinguish three forms of evidence. Scripted-rule comparisons measure behavior against specified testimony; adaptive crossplay tests responses to particular trained opponents; supplemental interventions investigate score and ballot mechanisms under explicit matching assumptions. None replaces the others. The hypothesis rule’s scripted-attack benefits coexist with truthful-testimony costs and targeted adaptive vulnerability.

Learning dynamics are also outcome-dependent. Multi-round defenders can suppress coalition victory while retaining frequent innocent ejections. The dependence-aware loop modestly lowers coalition victory after coalition stages yet increases whole-game voting harm. A smaller adaptation gap is therefore not itself a safety improvement or a stability certificate. The ten-generation sequence shows repeated gains against the latest opponent; fresh initialization, restricted policy classes, changing opponents, and finite evaluation prevent an asymptotic dynamical conclusion.

The Gaussian surrogate models corroboration capture as coalition-weight amplification. The game’s measured alibi errors occur while total coalition weight remains below its uniform share in the inspected scripted records. Alibi credit and the placement of self-incriminating penalties within downweighted rows offer different mechanisms, with attack-dependent trade-

offs under intervention. This distinction prevents the paper’s motivating analogy from becoming an unsupported explanation of every game outcome.

8 Limitations

1. **Restricted benchmark.** Small maps, structured claims, two policy-sharing impostors, and fixed observation/action spaces do not establish behavior in natural-language deliberation, independently incentivized coalitions, or deployed sensor systems.
2. **Finite learning and testing.** PPO stages start from fresh weights and have finite budgets. Crossplay contains a finite policy pool, not optimal responses over all possible strategies. The larger population has only three generations and five seeds.
3. **Inference resolution.** Five-seed exact tests cannot reject at raw 0.05, and larger ten-seed families encounter a corresponding Holm limit. Intervals are pointwise; Holm correction is within named families. Absence of rejection is not equivalence, and 303 specified contrasts do not make the original exploratory study prospectively preregistered.
4. **Information and hypothesis modeling.** Within-round histories are matched, but non-recurrent actors have limited cross-round memory. Hidden-partner controls change several information channels together. The hypothesis solver is exact for emitted constraints, with documented relaxations for some negative claims; normalized feasibility scores are not calibrated generative posteriors.
5. **Mechanism scope.** Matched initial evidence does not fix subsequent responses; ballot removal does not model behavioral adaptation. The bounded F2 replay does not estimate failure-route prevalence. Supplemental studies remain descriptive.
6. **Outcome definitions.** Creator quantities require incidents. Terminal coordination can be undefined after a prior coalition ejection. Whole-game harm includes incident-free meetings. Voting error, abstention, and coalition victory cannot be substituted for one another.
7. **Implementation and reproduction.** The audit changed several components together, so differences from historical experiments cannot be attributed to one isolated fix. Frozen source hashes, checkpoints, raw outcomes, and environment manifests support reproduction; different operating systems and PyTorch builds need not produce bit-identical training.
8. **Analytical correspondence.** The Gaussian surrogate has a strict reporter-minority condition and a parameter-restricted dependence result. Finite-sample weights and game testimony need not obey its amplification thresholds.

9 Conclusion

The completed corrected replication shows that defensive performance depends on the attacker, the training sequence, and the chosen outcome. Coalition learning increases harmful aligned voting without increasing directly false claims. Hypothesis scoring trades some scripted-attack accuracy gains for truthful-testimony costs and remains vulnerable to tar-

geted training. Repeated response learning produces renewed gains against the latest opponent over the observed horizon, without establishing an equilibrium or persistent cycle. Dependence-aware defenses can lower coalition victory while increasing innocent ejections. Combining explicit outcome definitions, adaptive crossplay, and bounded mechanism diagnostics yields a more qualified account of coalition deception than any single credibility or win-rate measure.

Acknowledgements

OpenAI Codex, using GPT-6, assisted with implementation review, correction and testing of research code, numerical reanalysis, and manuscript editing. Numerical measurements were produced by the executable experiments and analysis scripts described in this paper.

Data availability

Code, the manuscript, all final seed-level summaries, planned comparisons, and supplemental diagnostics are available in the public repository [Coalition Deception](#). The versioned corrected-study release supplies the original frozen source, retained checkpoints, raw episode and meeting records, and checksum manifests: [corrected-study-2026-09-11](#). The original and portable source fingerprints are distinguished in the reproduction guide. Appendix C specifies the analysis and manuscript build.

References

- [1] V. P. Crawford and J. Sobel, “Strategic information transmission,” *Econometrica*, vol. 50, no. 6, pp. 1431–1451, 1982, doi: [10.2307/1913390](#).
- [2] D. Lewis, *Convention: A philosophical study*. Harvard University Press, 1969.
- [3] E. Kharitonov, R. Chaabouni, D. Bouchacourt, and M. Baroni, “EGG: A toolkit for research on emergence of lanGuage in games,” in *Proceedings of the 2019 conference on empirical methods in natural language processing and the 9th international joint conference on natural language processing: System demonstrations*, 2019, pp. 55–60. doi: [10.18653/v1/D19-3010](#).
- [4] D. Condorelli and M. Furlan, “Cheap talking algorithms.” 2024. Available: <https://arxiv.org/abs/2310.07867v6>
- [5] K. Chen, “Communication with multiple senders.” 2025. Available: <https://arxiv.org/abs/2505.14639v1>
- [6] I. Arieli, I. Geffner, and M. Tennenholtz, “Resilient information aggregation,” in *Proceedings of the 19th conference on theoretical aspects of rationality and knowledge (TARK)*, in Electronic proceedings in theoretical computer science, vol. 379. 2023, pp. 31–45. doi: [10.4204/EPTCS.379.6](#).

- [7] M. Figura, Y. Lin, J. Liu, and V. Gupta, “Resilient consensus-based multi-agent reinforcement learning with function approximation.” 2021. Available: <https://arxiv.org/abs/2111.06776>
- [8] J. Witkowski and D. C. Parkes, “A robust Bayesian truth serum for small populations,” in *Proceedings of the 26th AAAI conference on artificial intelligence*, 2012, pp. 1492–1498. doi: [10.1609/aaai.v26i1.8261](https://doi.org/10.1609/aaai.v26i1.8261).
- [9] E.-M. El-Mhamdi, S. Farhadkhani, R. Guerraoui, and L.-N. Hoang, “On the strategyproofness of the geometric median,” in *Proceedings of the 26th international conference on artificial intelligence and statistics (AISTATS)*, in Proceedings of machine learning research, vol. 206. 2023, pp. 2603–2640. Available: <https://proceedings.mlr.press/v206/el-mhamdi23a.html>
- [10] K. Kopparapu *et al.*, “Hidden agenda: A social deduction game with diverse learned equilibria.” 2022. Available: <https://arxiv.org/abs/2201.01816>
- [11] B. Sarkar, W. Xia, C. K. Liu, and D. Sadigh, “Training language models for social deduction with multi-agent reinforcement learning,” in *Proceedings of the 24th international conference on autonomous agents and multiagent systems (AAMAS)*, 2025, pp. 1830–1839. doi: [10.65109/PHJJ9903](https://doi.org/10.65109/PHJJ9903).
- [12] J. Blumenkamp and A. Prorok, “The emergence of adversarial communication in multi-agent reinforcement learning,” in *Proceedings of the 2020 conference on robot learning*, in Proceedings of machine learning research, vol. 155. 2021, pp. 1394–1414. Available: <https://proceedings.mlr.press/v155/blumenkamp21a.html>
- [13] J. Li *et al.*, “Shapley counterfactual credits for multi-agent reinforcement learning,” in *Proceedings of the 27th ACM SIGKDD conference on knowledge discovery and data mining*, 2021, pp. 934–942. doi: [10.1145/3447548.3467420](https://doi.org/10.1145/3447548.3467420).
- [14] N. Orzan, E. Acar, D. Grossi, and R. Rădulescu, “Emergent cooperation under uncertain incentive alignment,” in *Proceedings of the 23rd international conference on autonomous agents and multiagent systems (AAMAS)*, 2024, pp. 1521–1530. Available: <https://arxiv.org/abs/2401.12646>
- [15] S. R. Motwani *et al.*, “Secret collusion among AI agents: Multi-agent deception via steganography,” in *Advances in neural information processing systems 37 (NeurIPS)*, 2024. doi: [10.52202/079017-2336](https://doi.org/10.52202/079017-2336).
- [16] F. Tramèr, N. Carlini, W. Brendel, and A. Madry, “On adaptive attacks to adversarial example defenses,” in *Advances in neural information processing systems*, 2020. Available: <https://arxiv.org/abs/2002.08347>
- [17] M. Lanctot *et al.*, “A unified game-theoretic approach to multiagent reinforcement learning,” in *Advances in neural information processing systems*, 2017. Available: <https://arxiv.org/abs/1711.00832>
- [18] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms.” 2017. Available: <https://arxiv.org/abs/1707.06347>
- [19] S. Huang and S. Ontañón, “A closer look at invalid action masking in policy gradient algorithms,” in *Proceedings of the international FLAIRS conference*, 2022. doi: [10.32473/flairs.v35i.130584](https://doi.org/10.32473/flairs.v35i.130584).

Appendix A. Proposition: corroboration capture under correlated testimony

Setting. Let $1 \leq m < n$, $N = n + m$, $f = m/N$, $\sigma, \sigma_c, h > 0$, $\gamma, \lambda \geq 0$, and $\rho \in [0, 1]$. Honest reports are $x_i = \theta + \varepsilon_i$ with independent $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$. Coalition reports are $y_j = \theta + b + \eta_j$, with $\eta \sim \mathcal{N}(0, \sigma_c^2[(1 - \rho)I + \rho \mathbf{1}\mathbf{1}^\top])$, independent of the honest noise. Define $r = (x, y)$, $c_i = \#\{k \neq i : |r_i - r_k| \leq h\}$, and the realized estimator $\hat{\theta}_{\text{cred}} = \sum_i w_i r_i$, $w_i \propto e^{\gamma c_i}$. Dependence-aware weights additionally multiply by $e^{-\lambda \text{ex}_i}$, where $\text{ex}_i = \max_{k \neq i} D_{ik} - (N - 1)^{-1} \sum_{k \neq i} D_{ik}$ and D_{ik} is the agreement frequency across T independent repetitions with fixed model parameters.

Exact agreement probabilities. With $s^2 = \sigma^2 + \sigma_c^2$,

$$p_{HH} = \text{erf}(h/(2\sigma)), \quad p_{CC}(\rho) = \text{erf}\left(\frac{h}{2\sigma_c\sqrt{1-\rho}}\right), \quad p_{CC}(1) = 1,$$

$$p_{HC} = \Phi((h - b)/s) - \Phi((-h - b)/s).$$

Hence $\bar{c}_H = (n - 1)p_{HH} + mp_{HC}$, $\bar{c}_C = (m - 1)p_{CC} + np_{HC}$ and $\Delta c = \bar{c}_C - \bar{c}_H$ is non-decreasing in ρ .

Deterministic mean-field surrogate. Replace each random corroboration count by its group expectation. The resulting coalition weight and estimator are

$$W_C = \frac{f}{f + (1 - f)e^{-\gamma\Delta c}}, \quad \tilde{\theta} = (1 - W_C)\bar{x} + W_C\bar{y}.$$

This substitution defines a surrogate. In general $W_C \neq \mathbb{E}[\sum_{j \in C} w_j]$, and it does not give an exact boundary for the realized estimator. Its exact bias and mean-squared error are

$$\text{bias}(\tilde{\theta}) = W_C b, \quad \text{MSE}(\tilde{\theta}) = W_C^2 b^2 + (1 - W_C)^2 \frac{\sigma^2}{n} + W_C^2 \sigma_c^2 \left(\rho + \frac{1 - \rho}{m} \right).$$

The approximation $\text{MSE}(\tilde{\theta}) \approx W_C^2 b^2$ requires the displayed variance terms to be negligible relative to $W_C^2 b^2$.

Proposition (surrogate amplification and robustness).

1. *Weight amplification.* For $\gamma > 0$, $W_C > f$ if and only if $\Delta c > 0$, equivalently

$$(m - 1)p_{CC} > (n - 1)p_{HH} - (n - m)p_{HC}.$$

At $\gamma = 0$, $W_C = f$. When $\Delta c > 0$, W_C increases with γ and is non-decreasing in ρ ; $W_C > 1/2$ precisely when $\gamma\Delta c > \log((1 - f)/f)$. Weight amplification increases the surrogate's squared bias for $b \neq 0$; a comparison of total MSE must also include the variance terms above. At $\rho = 1$ and negligible cross-group agreement, the amplification condition reduces to $(m - 1) > (n - 1)p_{HH}$. Negligible cross-group agreement requires separation relative to the noise scale, for example $(|b| - h)/s \gg 1$.

2. *Median robustness and the bias reference.* Because $m < n$, the pooled median M lies between the smallest and largest honest reports for every realization, irrespective of the coalition reports. Therefore

$$\text{MSE}(M) \leq \sigma^2 \mathbb{E} \left[\max_{1 \leq i \leq n} |Z_i|^2 \right] < \infty, \quad Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1),$$

uniformly in b and ρ . For odd N , writing $k = (N + 1)/2$ and $Z_{(j)}$ for honest order statistics gives the sharper pathwise bound $x_{(k-m)} \leq M \leq x_{(k)}$, hence $|\text{bias}(M)| \leq \sigma \mathbb{E}[Z_{(k)}]$. As n, m grow with $m/N \rightarrow f < 1/2$, the latter upper bound tends to $B_\infty = \sigma \Phi^{-1}(1/[2(1-f)])$. Thus $W_C|b| = B_\infty$ defines an asymptotic bias-comparison reference, not a finite-sample MSE break-even boundary. For $0 < W^* = B_\infty/|b| < 1$, $W_C|b| > B_\infty$ is equivalent to

$$\gamma \Delta c > \log \frac{(1-f)W^*}{f(1-W^*)}.$$

When $\Delta c > 0$ and the right-hand side is positive, division by Δc gives the positive reference gain γ^* . Finite-sample error comparisons use the actual risks or the finite-sample bound above. At $f = 1/2$, the median loses its uniform bound against arbitrarily displaced coalition reports; this does not imply that every estimator fails on every distribution.

3. *Dependence penalty.* In the long-window limit, for $m \geq 2$,

$$\begin{aligned} \bar{\text{ex}}_C &= \max(p_{CC}, p_{HC}) - \frac{(m-1)p_{CC} + np_{HC}}{N-1}, \\ \bar{\text{ex}}_H &= \max(p_{HH}, p_{HC}) - \frac{(n-1)p_{HH} + mp_{HC}}{N-1}. \end{aligned}$$

For $m = 1$, $\bar{\text{ex}}_C = 0$ and coalition correlation has no effect. The defended surrogate replaces $\gamma \Delta c$ by $\gamma \Delta c - \lambda(\bar{\text{ex}}_C - \bar{\text{ex}}_H)$. On any correlation interval where $m \geq 2$ and $p_{CC}(\rho) \geq p_{HC}$, its log-weight advantage depends on ρ through

$$\left[\gamma(m-1) - \lambda \frac{n}{N-1} \right] p_{CC}(\rho).$$

Consequently $\lambda \geq \gamma(m-1)(N-1)/n$ makes the defended surrogate weight non-increasing with correlation on that interval; a strict inequality makes it decrease where p_{CC} increases. For $5 + 2$, the threshold is 1.2γ . If $p_{CC} < p_{HC}$, the coefficient instead is $(m-1)[\gamma + \lambda/(N-1)]$, so the stated repair condition does not apply there. A comparison between $\rho = 0$ and $\rho = 1$ requires the first branch to hold throughout that interval.

Proof sketch. Groupwise softmax normalization gives W_C , and independence of the two groups gives the exact surrogate variance. The amplification and majority statements follow by rearranging the logistic expression. Fewer than half the reports are adversarial, so neither central rank can escape the honest range; for odd N at most m insertions can shift the

median's honest rank, giving the stated order-statistic bounds. Gaussian symmetry gives the bias bound, and convergence of central order statistics gives its fixed-fraction limit. Finally, the strong law applied to each pair's agreement indicators gives the long-window matrix entries. Taking each row's actual maximum and collecting coefficients in p_{CC} yields the two dependence branches. $\square$

The tests check the Gaussian probabilities, limiting concentration statistic, simulation identities, and boundary cases. Monte Carlo experiments also exhibit finite-sample reversals of the surrogate's weight-amplification prediction, so they do not establish exactness of its realized-estimator thresholds.

Appendix B. Configurations, hyperparameters, and scripted policies

Table B1. Corrected configurations and hyperparameters. All trained policies use completed-episode batches and independent evaluation streams.

Setting	Value
Rooms	Hub, Electrical, Navigation, Medbay
Map, meeting-only game	star: Hub–Electrical, Hub–Navigation, Hub–Medbay
Map, spatial game	star plus Electrical–Navigation
Free-play turns	6 (meeting-only), 8 (spatial)
Evidence, meeting-only game	random walk on the map, stay probability 0.35, one incident placed by the episode process
Evidence, spatial game	movement actions; two tasks per agent; incident requires one crew target and no other living crew in the room; the partner may be present; automatic report at the end of free play
Incident time	announced publicly by the report, so the scene window is the incident time step
Meeting	claim round (structured claim with polarity and confidence), response round, plurality vote with skip; ties do not eject; roles not revealed on ejection; random speaking order
Coalition	two impostors; partner identity in the observation; shared policy; shared reward; learned private symbol channel with four symbols (spatial game only)
Actor and critic	two-layer MLPs, 128 hidden units, non-recurrent
PPO	learning rate 3×10^{-4} , clip 0.2, $\gamma = 1$, GAE $\lambda = 0.95$, up to 8 epochs (target KL 0.05), 4 minibatches stratified by phase, value coefficient 0.5, entropy coefficient 0.02 annealed to 0.005, gradient clip 0.5
Updates per stage	400 standard; 800 or 1,600 for defender-budget controls
Generations (multi-generation loops)	10 (5+2 and 3+2), 3 (7+2), 4 (defended loop, single-meeting reward control), 2 (multi-round reward control)

Setting	Value
Rollout per update	64 completed episodes across 64 environments (F1); 32 completed episodes across 32 environments (spatial)
Evaluation	400 episodes every 20 updates (F1); 1,000 cross-play episodes per stage (F3, F4, controls); 1,000 per stage and 500 per cross-play cell (multi-generation)
Seeds	10 (F1–F4; F8 at 5+2; 5+2 multi-generation loops, defended loop and counterattack), 10 (3+2 loop), 5 (7+2 loop), 5 (F8 at 7+2, static learned-policy evaluations, controls); see Section 5 for evaluation-seed conventions
Credibility temperatures	1.5 (soft), 0.15 (sharp)
Dependence-aware defense	$\lambda = 4$, window of three meetings, top-1 partner
Hypothesis crew	honesty probability 1/2, ejection threshold 1/2, zero weight for impossible hypotheses; no pseudoweight

Suspicion weights of the scripted crew. Own evidence: an agent the voter saw in the incident room during the scene window +3.0; an agent the voter saw elsewhere −3.5; an agent the voter could not place +1.0; a speaker with no informative claim +0.4. Testimony: a claim the voter personally knows to be false +4.0 on its speaker (the caught-lying penalty); a publicly self-contradictory claim +3.0 on its speaker; a claim placing an agent in the incident room during the window +1.5; a claim by another speaker placing an agent elsewhere during the window −2.0 (the alibi-witness term); a speaker’s own claim placing itself elsewhere −0.5; a positive was-with claim −1.5 on the subject and −0.5 on the speaker; an accusation +0.8 on its target, a contradiction +0.4. The vote threshold is 0.5. Credibility is −3.0 when the voter catches the speaker lying; otherwise −2.0 for a self-contradictory claim set; otherwise the count of other informative speakers whose claims are jointly satisfiable. The caught and contradictory branches replace the corroboration count.

Scripted coalitions. The full behavioral definitions are in Section 3.3. In particular, alibi support is asymmetric (creator self-location plus partner sighting/defense), and the framer makes scene-placement claims rather than necessarily truthful testimony. Spatial movement uses the script’s own sightings and other legitimately available information.

Appendix C. Reproduction

The public package separates code, paper, and final analysis data. The original completed study identifies its 82-file executable/configuration snapshot by SHA-256 691f7687435424185064b24a9fee1832c6e7b32b9661efe20a10e35ca0c17d35. The cleaned

portable package has a different source fingerprint, recorded in its packaging provenance, while preserving all 240 job designs and the scientific engine. Original run outputs retain the original protocol; they are not relabeled as products of the portable tree.

The study completed 1,255 learner stages and retains 11,885 checkpoints, including a genuinely untrained F1 checkpoint, intermediate checkpoints every 20 F1 or 50 spatial updates, and final checkpoints. Validation checks restricted deserialization, finite architecture-compatible tensors, expected update counts, and equality of final versus final-numbered parameters. It checks raw episode and meeting outcomes against aggregates, exact seed cohorts, every planned comparison, source identity, and configuration. A completed history file alone does not mark an experiment complete.

The reproducibility archive supplies the frozen engine and protocols, raw main-study and supplemental data, environment manifests, current analysis/verification tools, and checksums. The current analyzer and renderer are separate from the immutable engine snapshot. The final analysis contains all 303 contrasts in 15 families; the manuscript presents selected estimates, while the complete tables retain every contrast, metric, and seed identity. The finite-pool gap file includes the selected policies and conservative within-matrix uncertainty bands.

The mechanism supplement includes 30 jobs, 540,000 episodes, and 240,000 honest-voter records. The ballot supplement includes 15 trained input jobs, 135 evaluation cells, 67,500 episodes, and 317,670 honest score views. Source, checkpoint, reset, and raw-record identities are retained. Independent verification recomputes tally necessity/sufficiency, abstention, score-derived flags, and aggregate summaries; it does not turn feasibility scores into exact probabilities.

The self-contained paper build converts the canonical Markdown into TeX and PDF using Pandoc, the supplied caption filter, and XeLaTeX. Its final tables are selected from validated machine-readable summaries; the paper directory retains a source-value manifest. The release README provides exact analysis, checksum-verification, extraction, and build commands. Historical drafts, incomplete jobs, smoke outputs, and superseded analyses are not used as evidence for the results in this manuscript.